

A
DOCTOR
ENJOYS
SHERLOCK
HOLMES

Edward J. Van Liere



VANTAGE PRESS NEW YORK WASHINGTON HOLLYWOOD

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ENJOYS SHERLOCK HOLMES ***

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HOLMES

“Nor must you find fault with me if I often give you what I have borrowed from my various reading, in the very words of the authors themselves” (Macrobius—trans. by Boswell).

Boswell: *The Hypochondriack*, No. XXI

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Edward J. Van Liere



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FIRST EDITION

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To my grandsons:

Edward Van Liere Batchelder and Terry Lewis Batchelder.

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To undertake the writing of a large book is like entering on a long and difficult journey, in the course of which much fatigue and uneasiness must be undergone, while at the same time one is uncertain of reaching the end of it: whereas writing a short essay is like taking a pleasant airing that enlivens and invigorates by the exercise which it yields, while the design is gratified in its completion.

Boswell: *The Hypochondriack*, No. I

DOCTOR WATSON AND THE WEATHER

“It had been a close and rainy day in October.”

The Resident Patient

To me one of the most delightful touches in the tales of Sherlock Holmes is the frequent mention of the state of the weather. As far as I know, no one has emphasized the numerous references to the weather by Dr. Watson. Especially in the opening of the short stories can these be found, but they are not confined there, for they appear throughout the longer tales as well. Many instances can be cited to illustrate Dr. Watson’s allusions to the state of the weather. Let us freshen our memory by reviewing some of these.

Numerous references to the rains and winds of autumn may be found, such as, “... the weather had taken a sudden turn to rain, with autumnal winds” (*The Adventure of the Noble Bachelor*). And, in a similar vein, “Outside the wind still screamed and the rain splashed and pattered against the windows. This strange, wild story seemed to have come to us from amid the mad elements—blown in upon us like a sheet of seaweed in a gale” (*The Five Orange Pips*). And in *The Problem of Thor Bridge*, we find that, as Dr. Watson was dressing one morning, he observed how the leaves were being whisked away from the plane tree (we would call it a sycamore) which graced their back yard.

The events related in *The Hound of the Baskervilles* took place in the autumn of the year, and several interesting allusions to this season may be found: “I walked over to the black window, and I looked through a blurred pane at the driving clouds and at the tossing outline of the wind-swept trees. It was a wild night....” Another reference, “A dull and foggy day with a drizzle of rain. The house is banked in with rolling clouds....” Also, “All day today the rain poured down, rustling on the ivy and dripping from the eaves....” And in another instance, “Rain squalls drifted across ... and the heavy, slate-coloured clouds hung low over the landscape, trailing in gray

wreaths down the sides of the fantastic hills.” And lastly, an especially interesting allusion may be cited:

We hurried through the dark shrubbery, amid the dull moaning of the autumn wind and the rustle of the falling leaves. The night air was heavy with the smell of damp and decay. Now and again the moon peeped out for an instant, but clouds were driving over the face of the sky, and just as we came out on the moor a thin rain began to fall.

The paragraph just quoted pictures a rather wet autumn night, but not an especially bad one. In *The Adventure of the Golden Pince-Nez*, an extremely wild and stormy night toward the end of November is depicted. Dr. Watson describes vividly how the wind howled down Baker Street, and how the rain beat vigorously against the windows. Another reference may be cited which depicts gloomy autumn days: “It had been a close and rainy day in October....” (*The Resident Patient*).

It would appear, then, that many of Sherlock Holmes’ adventures took place during the fall of the year. It must be inferred that the dark and stormy days of autumn depressed the great detective, for we find him saying to Dr. Watson, “Draw your chair up and hand me my violin, for the only problem we still have to solve is how to while away these bleak autumnal evenings” (*The Adventure of the Noble Bachelor*).

From the instances just enumerated, it would seem that Dr. Watson was fond of picturing a background of wind, rain, and storm for many of his stories. He also described the well-known fogs of London with telling effect. In *The Adventure of the Copper Beeches*, Dr. Watson writes, “A thick fog rolled down the lines of dun-coloured houses, and the opposing windows loomed like dark, shapeless blurs through the heavy yellow wreaths.” And again, “It was a September evening and not yet 7 o’clock, but the day had been a dreary one, and a dense drizzly fog lay low upon the great city” (*The Sign of the Four*). In the same story, we find Holmes saying, “See how the yellow fog swirls down the street and drifts across the dun-coloured houses.” In *The Adventure of the Bruce-Partington Plans*, Dr. Watson tells how, during the month of November of 1895, such a heavy fog descended on London that it was impossible to see the houses on the opposite side of Baker Street.

In addition to wind, rain, storm, and fog, Dr. Watson makes reference as well to sharp winter weather. In *The Adventure of the Blue Carbuncle*, it will be remembered that Dr. Watson visited Sherlock Holmes two days following Christmas to give his friend the season's greetings. Watson writes, "I seated myself in his arm chair and warmed my hands before his crackling fire, for a sharp frost had set in, and the windows were thick with ice crystals." When he and Holmes set out to obtain information about the goose which had swallowed the blue carbuncle, Watson tells us: "It was a bitter night, so we drew on our ulsters and wrapped cravats about our throats. Outside, the stars were shining coldly in a cloudless sky, and breath of the passers-by blew out into smoke like so many pistol shots. Our footfalls rang out crisply and loudly...."

In *The Adventure of the Abbey Grange*, Watson describes an adventure which took place on a bitterly cold morning during the winter of '97. On another occasion, Watson tells how he and Holmes went for a walk on a cold and frosty winter evening (*The Adventure of Charles Augustus Milverton*). A description of a beautiful winter day may be found in *The Adventure of the Beryl Coronet*: "It was a bright, crisp February morning, and the snow of the day before still lay deep upon the ground, shimmering brightly in the wintry sun."

Mention is made too of the weather in the spring. On several occasions, references are made to the cold and stormy weather of the early part of this season: "It was a cold morning of the early spring, and we sat after breakfast on either side of a cheery fire in the old room at Baker Street" (*The Adventure of the Copper Beeches*). *The Adventure of the Speckled Band* took place in the spring of '83. Watson writes: "It was a wild night. The wind was howling outside, and the rain was beating and splashing against the window." In *The Adventure of the Wisteria Lodge*, the good doctor writes that it was a cold, dark, windy March evening, and a fine rain was falling. And in *His Last Bow*, Watson writes that he had recorded in his notebook that, in the latter part of March in the year 1892, it was a bleak and windy day.

Milder spring weather also claimed Dr. Watson's attention: "It was an ideal spring day, a light blue sky, flecked with little fleecy clouds drifting across from west to east. The sun was shining very brightly, and yet there was an exhilarating nip in the air, which set an edge to a man's energy"

(*The Adventure of the Copper Beeches*). In a similar vein, “It was a perfect day, with a bright sun and a few fleecy clouds in the heavens” (*The Adventure of the Speckled Band*). In *The Adventure of the Three Garridebs*, a lovely spring evening is mentioned, and Watson writes that the slanting rays of the setting sun made even a little prosaic street look golden.

For some reason or other, less frequent mention is made of summer weather. There are, however, several interesting references to this enjoyable season. In the story of *The Greek Interpreter*, Dr. Watson writes, “It was after tea on a summer evening....” And similarly, in *The Adventure of the Mazarin Stone*, our attention is called to the fact that it was the evening of a lovely summer’s day. It would be expected that somewhere in the stories an allusion would be made to the torrid days of August: “It was a blazing hot day in August. Baker Street was like an oven, and the glare of the sunlight upon the yellow brickwork of the house across the street was painful to the eye” (*The Adventure of the Cardboard Box*). An especially poignant reference to a hot summer night may be found in the story *His Last Bow*. Dr. Watson gives a vivid description of a hot night on the second of August. He mentions that there was an awesome hush and a feeling of expectancy in the sultry and stagnant air. The August to which reference is made was that of 1914—the beginning of World War I. In that holocaust, Great Britain lost the flower of her youth. She has not yet recovered from that mortal blow.

People living in the Victorian era, just like many of us today, were interested in barometric pressure. In *The Boscombe Valley Mystery*, an interesting reference to barometric readings may be found. Sherlock Holmes remarks to Lestrade of Scotland Yard: “How is the glass? Twenty-nine, I see. No wind, and not a cloud in the sky.” Somewhat later in the day, he mentions to Dr. Watson: “The glass still keeps very high.... It is of importance that it should not rain before we are able to go over the ground.” Fortunately, in this instance good weather prevailed, for Watson writes, “There was no rain, as Holmes had foretold, and the morning broke bright and cloudless.”

One might well ask why Dr. Watson emphasizes the state of the weather so often? Several reasons come to mind: To make the setting of his stories more realistic; to take up space in the manuscript; to use the subject of the weather as an excuse for some fine writing; to reveal Watson’s little-

suspected love of nature; or perhaps still other reasons. Let us examine some of these.

It is well known that writers often have dirty weather prevailing when foul deeds are to be committed. Dr. Watson, however, does not abuse this privilege in his writings, although he does occasionally take advantage of unusual atmospheric conditions to make the story more exciting. A good example of this is to be found in the novel *The Hound of the Baskervilles*.

One recalls that Sherlock Holmes laid a trap to catch the villain, Stapleton. A part of the plan made it necessary for the hero, Sir Henry, to walk across the dismal moor alone after dark. Holmes had calculated that Stapleton would let loose his spectral hound that particular night. It was carefully arranged, of course, so that Sherlock Holmes, Dr. Watson, and Lestrade would kill the hound before any harm befell Sir Henry, and then catch Stapleton redhanded. All went according to plan until a dense fog started to rise from the adjacent Grimpen Mire. The low visibility produced by the unexpected fog almost upset Holmes' well-laid plans and nearly caused the death of Sir Henry, because his friends could not see the hound until it was dangerously close to him.

Can the good doctor be accused of writing about the weather simply to make the manuscripts longer? I think not. In the first place, there is no evidence (certainly not in his early career) that the money he received for his stories bore any relation to their length. Further, and more important, a man with Watson's imagination did not have to rely upon such a mundane topic as the weather to lengthen his stories. We can dismiss this argument promptly by stating that he was not guilty of padding his manuscripts in this manner.

Did he use the weather as an occasional excuse for fine writing? Probably not, although some of the paragraphs in which he describes the state of the weather, if examples of fine writing, are certainly examples of vivid writing as well. Several instances could be given, but one will suffice:

It was in the latter days of September, and the equinoctial gales had set in with exceptional violence. All day the wind had screamed and the rain had beaten against the windows, so that even here in the heart of great, handmade London we were forced to raise our minds for the instant from the routine of life,

and to recognize the presence of those great elemental forces which shriek at mankind through the bars of civilization, like untamed beasts in a cage. As evening drew in, the storm grew higher and louder, and the wind cried and sobbed like a child in the chimney.

The Five Orange Pips

What other reasons could be given for Watson's frequent allusions to the weather? There are, of course, the dapper gentleman and the Scotchman, who are conscious of the state of the weather since they deplore—for different reasons, to be sure—losing the press in their trousers by being caught in the rain. To the Londoner, however, who seems obviously unaware of the somewhat disheveled state of his attire, such an explanation could scarcely apply. On the other hand, one cogent reason does come to mind. It must always be remembered that Watson was both a physician *and* an author. Now the state of the weather is of perpetual interest to a general practitioner of medicine, and this is what Dr. Watson pretended to be. It is true that the physician who practices in the city probably is not as weather-minded as his colleague in the country. Dr. Watson lived in the great city of London, but often made his professional calls on foot and doubtless trudged through storm and rain: "When your round is a short one you walk, and when it is a long one you use a hansom" (*The Crooked Man*). He mentions that after his marriage he dropped in at 221B Baker Street from time to time when walking in that vicinity.

In point of fact, the kind of weather is of no little importance to a person who is called out at any time of day or night, as almost any physician will testify. This may be one of the important reasons why Watson makes such frequent mention of the weather in his stories.

All may not agree with me that such descriptions of the state of the weather add greatly to the interest of Watson's writings. This is a matter of opinion; for doubtless there are many men who have little interest in the weather under any circumstances. Earl Derr Biggers, the creator of Charlie Chan, had one of his less liked characters remark that he was not interested in the weather, since he was not a cabbage. Be that as it may, I have always pitied those individuals who pay absolutely no attention to the weather; they are missing a great deal in life.

There are people who see beauty and interest only in a beautiful day. The poet probably was right when he sang, “What is so rare as a day in June?” But would anyone deny that there is beauty in a snowstorm or a sleetstorm, and is not a cloudburst or a thunderstorm an awesome spectacle, and can not even a duststorm be a striking phenomenon? Now nature, I grant, can overdo this matter of storms, and too much dust in the atmosphere or a week of rainy, sullen weather leave much to be desired.

To those of us who have been thrilled by the saga of Sherlock Holmes and Dr. Watson, and who have found enjoyment and relaxation in reading about these famous characters, I would like to make one further suggestion: Let us think of Dr. Watson not only as a practitioner of medicine, or only just as an amateur detective and confidant of Sherlock Holmes, but also as a man interested in many things—a lover of nature and one who could see charm in all her moods.

THE ANATOMICAL SHERLOCK HOLMES

“I believe he is well up in anatomy....”

A Study in Scarlet

A number of references to the anatomy laboratory and to portions of the cadaver may be found in the tales. Shortly after Watson's return from India, he met his old friend Stamford at the Criterion bar in London. This was a memorable occasion, for it was Stamford who introduced Watson to Sherlock Holmes. When Stamford was telling Watson something about Holmes, he said, “I believe he is well up in anatomy....” Stamford made it clear, however, that Holmes was not a medical student, but did have pronounced scientific interests. He emphasized the fact that Holmes had a cold scientific approach to problems which deeply interested him and that he had been seen “beating the subjects in the dissecting room with a stick....” When Watson evinced some surprise at this unbecoming behavior, Stamford explained, “to verify how bruises may be produced after death” (*A Study in Scarlet*).

At first glance, Holmes' activity seems to be inexcusable and betokens a deplorable disrespect for the dead. His researches must not be regarded in this light, however, for it may be of distinct medicolegal interest to ascertain whether the bruises on a body were produced before or after death. This problem has been studied extensively, and information concerning it may be found in books dealing with medicolegal matters. Holmes' early interest and study in this field are quite understandable, for at that time he was laying the foundation for his brilliant career as a specialist in the study of crime.

When Watson first started rooming with Holmes he found that the latter “Sometimes ... spent his day ... in the dissecting rooms....” (*A Study in Scarlet*). In one instance, reference is made to the preservation of bodies in the anatomy laboratory: “Bodies in the dissecting rooms are injected with preservative fluids.” The agents employed in the embalming fluid are given:

“carbolic, or rectified spirits would be the preservative....” (*The Adventure of the Cardboard Box*).

Osteology must have appealed to Holmes, since bones are frequently mentioned. Once when Holmes was trying to analyze a difficult case, he made allusion to Cuvier, the famous anatomist and anthropologist: “Cuvier could correctly describe a whole animal by the contemplation of a single bone....” (*The Five Orange Pips*). In *A Study in Scarlet*, a description of the alkali plains of our great West is given; these pointed but gruesome sentences may be found:

Here and there are scattered white objects which glisten in the sun and stand out against the dull deposit of alkali. Approach and examine them. They are bones: some large and coarse, others smaller and more delicate. The former have belonged to oxen and the latter to men.

In another instance, Holmes was consulted about a “charred fragment of bone” which had been recovered from a heating furnace in a private dwelling. He showed it to Watson, who without hesitation stated that it was the upper condyle of a human femur (*The Adventure of Shoscombe Old Place*).

One early spring afternoon, Holmes and Watson upon returning from a walk found that a visitor had left his pipe on the table. Holmes picked it up and examined it carefully. Watson writes, “He held it up and tapped on it with his long, thin forefinger, as a professor might who was lecturing on a bone” (*The Yellow Face*). This is an apt allusion, for many medical students will remember the occasion when an instructor in anatomy held a bone in his hand, pointed out areas where muscles had been attached, and commented on other characteristic features. One is reminded of the story of the poet-physician Oliver Wendell Holmes, who was professor of anatomy at Harvard for many years. Once when lecturing on the sphenoid bone (a bone of the skull having an exceedingly complicated architecture), he is reputed to have said something like this: “Gentlemen, I have in my hand the sphenoid bone. Gentlemen, I say d—— the sphenoid bone!” Any medical student who has studied this bone will wholeheartedly agree with the remark.

Mention is made also of fossil bones. When Holmes and Watson visited the house of Nathan Garrideb, they noticed a large cupboard full of them. Holmes was invited to take a seat, but his host found it necessary to clear the chair of bones. Obviously, Garrideb was a most enthusiastic collector (*The Adventure of the Three Garridebs*).

Sherlock Holmes presumably enjoyed anthropology, for several references to this science may be found in the tales. In *The Hound of the Baskervilles*, when Dr. Mortimer first met Sherlock Holmes, the doctor rather facetiously said to him:

“I hardly expected so dolichocephalic a skull or such well-marked supra-orbital development. Would you have any objection to my running my finger along your parietal fissure? A cast of your skull, sir, until the original is available, would be an ornament to any anthropological museum. It is not my intention to be fulsome, but I confess that I covet your skull.”

Holmes apparently was slightly annoyed at this frank but somewhat insensate disquisition, and remarked, “You are an enthusiast in your line of thought, I perceive, sir, as I am in mine.”

On another occasion when Dr. Mortimer was speaking of Sir Henry Baskerville, he stated: “A glance at our friend here reveals the rounded head of the Celt, which carries inside it the Celtic enthusiasm and power of attachment. Poor Charles’ head was of a very rare type, half-Gaelic, half-Ivernian in its characteristics.”

Dr. Mortimer obviously was a keen observer, a person of scholarly tastes and “a most learned man in his own line.” When he was telling Sherlock Holmes and Dr. Watson of his friendship with Sir Charles Baskerville, he remarked, “... and many a charming evening we have spent together discussing the comparative anatomy of the Bushman and the Hottentot.”

Interest in anthropology is evinced further by mention of prehistoric man. One day, while Dr. Watson was walking on the moor, he met the naturalist Stapleton, the villain in *The Hound of the Baskervilles*. Watson’s attention was attracted to the circular rings of stone on a hillside. He asked his companion whether they were the ruins of ancient sheep pens. Stapleton

replied, “Prehistoric man lived thickly on the moor....” When Watson questioned him as to when the moor was inhabited; the answer was, “Neolithic man—no date.”

Another allusion to ancient man may be cited. In the house of Garrideb, as mentioned, Holmes and Watson noticed above a cupboard a series of plaster skulls; the names “Neanderthal,” “Heidelberg,” and “Cromagnon” were printed underneath them (*The Adventure of the Three Garridebs*). Not only the anthropologist but the trained biologist as well is, of course, quite familiar with the names of our early ancestors.

Once Holmes’ knowledge of anthropology helped him solve an important case. A maiden lady had received in the mail a small cardboard box containing two human ears—one that of a woman. Holmes was called in, and during the course of his investigations visited the receiver of the gruesome package. He noted the resemblance of one of the severed ears to those of the lady upon whom he was calling. Because of the striking likeness, he felt certain that the person whose ear had been dismembered was a close relative of the lady he had come to interview.

Somewhat later, he gave a lecture—a thing he was prone to do—to Dr. Watson on the surface anatomy of the ear:

... there is no part of the body which varies so much as the human ear. Each ear is as a rule quite distinctive and differs from all others.... I ... examined the ears in the box with the eyes of an expert and had carefully noted their anatomical peculiarities.... I perceived that her [Miss Cushing’s] ear corresponded exactly with the female ear I had just inspected. The matter was entirely beyond coincidence. There was the same shortening of the pinna, the same broad curve of the upper lobe, the same convolution of the inner cartilage. In all essentials it was the same ear.

Of course, I at once saw the enormous importance of the observation. It was evident that the victim was a blood relative, and probably a very close one.

The Adventure of the Cardboard Box

This was an exceedingly astute observation on Holmes' part. He was right when he remarked that as a rule each ear is quite distinctive, and cleverly pointed out the important anatomical features. In any event, the study he made of Miss Cushing's ear aided him greatly in solving the mystery of the cardboard box, and we know that the murderer was promptly apprehended.

In the story just related, a newspaper of the day was supposed to have suggested that the preserved ears had been sent by medical students as a joke. It appears that the maiden lady at one time had had unpleasant dealings with medical students: "... she let her apartments in her house to three young medical students, whom she was obliged to get rid of on account of their noisy and irregular habits.... [The ears were sent] by those students who owed her a grudge and who hoped to frighten her by sending her those relics of the dissecting rooms."

Although this theory was later proved to be false, it was not too farfetched. Indeed, many, many stories can be told about the behavior of medical students in the anatomy laboratory. A favorite minor prank, for example, is to cut off a finger or an ear and slip it into the pocket of an unsuspecting visitor. This bit of horseplay probably discourages future visits to the anatomical laboratory. Although anatomy instructors deplore such practices, they are likely to overlook them, because laymen are not encouraged to visit dissecting rooms.

In the opening of one of the stories, we find Holmes stooping over a low-power microscope (*The Adventure of Shoscombe Old Place*). He explained to Watson that there were epithelial cells in the microscopic field. As far as I am aware, this is the only reference to individual body cells to be found in the tales. It appears, then, that Holmes was much more interested in gross structures of the body, especially osteology, than in microscopic structures.

We have seen that many pertinent allusions to anatomical science may be found in the tales. In one instance, at least, Holmes' intimate knowledge of surface anatomy—that is, the configuration of the external ear—enabled him to solve handily a perplexing mystery. The allusions made to anatomical matters are of especial delight to those of us in the field of biology.

“BRAIN FEVER” AND SHERLOCK HOLMES

“I have only just recovered from nine weeks of brain fever and am still exceedingly weak.”

The Naval Treaty

Several years ago, there appeared in the *Journal of the American Medical Association* a provocative article with the arresting title, “Brain Fever,” written by Louis Cassamajor.^[1] The author is to be commended for his courage in choosing this unusual title, and the *Journal* to be congratulated for publishing it. Although I welcome the term “brain fever,” I did rub my eyes, because I had not seen it employed for a long time. In my youth the term was used commonly, but was dismissed from my mind when I commenced the study of medicine. It was relegated to the same limbo as the old expression “typhoid malaria.”

In order that I may develop my thesis, a brief review of Cassamajor’s article is in order. The author points out that, in the early part of the past century, considerable literature appeared describing a disease known as “brain fever” (called also “hydrocephalic fever” and sometimes “encephalitis”). For the main part, it occurred in children. The illness subsided after a few days to a couple of weeks, and the patient usually recovered.

The author brings out further that, although the disease was apparently accompanied by fever, there are no recorded temperatures, for the modern clinical thermometer was not invented until 1868. It is emphasized, also, that no neurological signs appeared in the case reports. It was only after the writings of Erb and of Westphal in 1875 that neurological examination, as we now know it, began to develop. For some unknown reason, about 1850 mention of the disease disappeared from medical literature. The author, however, makes the statement: “Undoubtedly the condition does exist today.”

Following a brief historical introduction, the author gives in some detail the case histories of four children, the youngest six and one-half and the

oldest eleven years of age, whom he had rather recently diagnosed as suffering from “brain fever.” It is highly gratifying that they all made a complete recovery. The disease is characterized by signs and symptoms indicating a considerable brain involvement, “including convulsions, comas, paralyses, cerebellar asynergy and a sort of bulbar palsy.” The onset is irregular, except when head trauma has been previously sustained, when it may be sudden.

One reason, among others, why this stimulating article especially interested me was that the term “brain fever” called to my mind the immortal stories of Sherlock Holmes. In them several individuals are described as suffering from this condition. It is of nostalgic interest to examine the circumstances which surrounded these victims when they were stricken.

We find in one of the stories that a housemaid, with an unstable Celtic temperament, “had a sharp touch of brain fever.” She had had a violent love affair with a handsome but perfidious butler, who had thrown her over for another girl. Following her partial recovery, she had taken a terrible vengeance and was directly responsible for her faithless lover’s death. When questioned about him by the master of the household, she became hysterical and unmanageable: “For two days [she] had been so ill, sometimes delirious, sometimes hysterical...” (*The Musgrave Ritual*). She evidently made a rapid recovery, for on the third night she disappeared and her whereabouts were never discovered.

In another story, a young girl whose mother had died was treated cruelly by her father, who had remarried. She had an income of her own which she generously allowed her father to use. When she fell in love with a young man, her father tried desperately to make her sign a contract providing that, in the event of marriage, he could still use her money. This she refused to do. He placed her in solitary confinement so that she could not see her lover, and treated her inhumanly in other ways: “... he kept on worrying her until she got brain fever, and for several weeks was at death’s door” (*The Adventure of the Copper Beeches*). It is pleasant to relate that she recovered and succeeded in eloping with her lover and presumably lived happily ever after.

In still another story, in which Sherlock Holmes attempted to gain an audience with a middle-aged spinster, he was informed that she was too ill to be interviewed. Her doctor said: "She has been suffering since yesterday from brain symptoms of great severity. As her medical adviser, I cannot possibly take the responsibility of allowing anyone to see her. I should recommend you to call again in ten days" (*The Adventure of the Cardboard Box*). This illness, somewhat later in the story, is referred to specifically as "brain fever."

The illness had developed when this spinster heard that conscientious diplomat. Through his own carelessness, how-her younger sister had been foully murdered by her husband. It was the spinster who, by wicked machinations, had been largely responsible for her sister's death. It is noteworthy that Holmes was advised to come back ten days later. This indicates that the disease was not of long duration.

One of Dr. Watson's former schoolfellows, Percy Phelps, wrote him, "I have only just recovered from nine weeks of brain fever and am still exceedingly weak." In his letter, he further informed Dr. Watson that he wished to consult with Sherlock Holmes, and asked his friend to bring him, since he was in deep trouble. This able young man was an earnest and ever, he had lost an important state document. In narrating his story to Sherlock Holmes and Dr. Watson, the patient gave a vivid description of his distressing illness: "Here I have lain, Mr. Holmes, for over nine weeks, unconscious, and raving with brain fever ... in my mad fits I was capable of anything. Slowly my reason has cleared, but it is only during the last three days that my memory has quite returned" (*The Naval Treaty*).

This poor fellow had an illness of long duration and, according to the story, his strength came back but slowly. It will be recalled that Sherlock Holmes solved the mystery and was able to place the important state document again in the patient's hands. As far as we know, he eventually made a complete recovery in spite of the severity of the attack.

In *The Hound of the Baskervilles*, it will be remembered that the heir, Sir Henry Baskerville, narrowly escaped death on the moor. The shock of the adventure with the hound, coupled with the fact that the woman with whom the heir was in love was actually the wife of the villainous Stapleton, his would-be murderer, was too much for our hero. Watson writes: "But the

shock of the night's adventure had shattered his nerves, and before morning he lay delirious in a high fever under the care of Dr. Mortimer. The two of them were destined to travel together round the world before Sir Henry had become once more the hale, hearty man that he had been before he became master of that ill-omened estate."

In this last instance, Watson does not specifically state that the patient was afflicted with "brain fever," but the implication is plainly there. It is, moreover, worthy of note that it took the victim a long time to regain his health.

The individuals whom Dr. Watson described as suffering from "brain fever" obviously had all passed through a terrific mental storm—in modern parlance, they had sustained "severe psychic traumata." Whether this alone could cause "brain fever" is a moot question. The condition presumably is caused by a virus. That a severe brain storm could cause the lurking virus to become active seems unlikely.

Previously it has been mentioned that a head injury apparently is capable of hastening the onset of brain fever in children. (We will waive the fact that the cases Dr. Watson described are adults.) A physical injury presumably produces certain organic changes in the brain or in its meninges, which perhaps could precipitate an attack of encephalitis, assuming that the virus was present. There is no particular evidence that the patients mentioned in the tales had suffered a head injury.

It is true, of course, that great emotional upsets are often accompanied by marked vascular disturbances. These may manifest themselves in the brain as well as in other parts of the body. It is not conceivable that encephalitis would follow an emotional storm. Be that as it may, this can be said: It is generally agreed that anything which lowers the resistance of an individual may make him more susceptible to disease. In the cases described by Dr. Watson, the virus may have been present, and the shock produced by the emotional storm, which all these people experienced, might have precipitated an attack of encephalitis (brain fever).

If this seems too farfetched, and the reader cannot go along with me, I can say only that I am sorry. I will have to use the argument that one should allow a talented and imaginative writer like Dr. Watson a liberal degree of poetic license.

Dr. Watson may have used the term “brain fever” loosely, and perhaps as synonymous with extreme nervous exhaustion. He does, however, mention that some of the sufferers became delirious. This symptom is suggestive of encephalitis. Also, in one or two instances the victim recovered rather quickly, which points to an acute condition such as encephalitis rather than nervous exhaustion. I am cognizant that some may regard this as a specious argument.

The question could be raised whether the patients described by Dr. Watson suffered from hysteria. This condition cannot entirely be ruled out. Hysteria has protean manifestations, and may even be accompanied by fever. The great mental storms through which these patients passed are conducive to hysterical attacks. The fact, however, that these individuals ran a high fever and were dangerously ill would militate against a diagnosis of hysteria. It seems fairly safe to assume that the disease from which they suffered probably had an organic basis.

The reader should be reminded that Dr. Watson began the study of medicine only a short time after the researches of Erb and of Westphal on the nervous system had been published. Neurology had not yet come into its own, and it is likely that the medical profession did not make fine distinctions when dealing with diseases of the brain or its meninges. It probably is not charitable for me to suggest that the professors who taught the young Watson about nervous diseases in the year 1876 or thereabouts had not kept up with the literature in their field—a fault of which we are all more or less guilty.

Dr. Watson has been taken to task by some critics in the medical profession for using the term “brain fever,” and the implication has been made that his employment of a meaningless term was unworthy of a medically trained man. Now it appears that this criticism is unjustified. We might quarrel with Dr. Watson as to what brought on the attacks of “brain fever,” or whether the victims actually had the disease in the cases he so vividly described, but the term itself is acceptable.

The Holmesian enthusiast will rejoice that the term “brain fever” is again in good repute and is accepted by the medical fraternity. The very fact that the staid *Journal of the American Medical Association* has published an article bearing the title “Brain Fever” has stripped criticism of all weapons.

The loyal Holmesian no longer needs to feel apologetic for the nomenclature Dr. Watson used to describe a rather unusual and fortunately rare clinical entity.

CURARE AND SHERLOCK HOLMES

“... for the action of the alkaloid is rapid.”

A Study in Scarlet

Curare is used as a lethal agent in two of the Sherlock Holmes stories. In the novel *A Study in Scarlet*, liberties are taken with the pharmacologic properties of curare, for actions are ascribed to this agent which it could not possibly have had. But in *The Adventure of the Sussex Vampire*, curare is employed scientifically and with telling effect; and indeed, an interesting plot is built around this agent.

Let us first examine the role curare played in *A Study in Scarlet*. Holmes, it will be remembered, wished to determine the toxicity of certain pills and instructed Dr. Watson to fetch a little dog, which already was *in extremis*: “... that poor little devil of a terrier which has been bad so long, and which the landlady wanted you to put out of its pain yesterday.” Watson commented on the state of the animal: “Its laboured breathing and glazing eye showed that it was not far from its end. Indeed, its snow-white muzzle proclaimed that it had already exceeded the usual term of its existence.” Then, according to Dr. Watson, some of the pills which were thought to contain curare were dissolved in milk and offered to the sick dog: “The unfortunate creature’s tongue seemed hardly to have been moistened in it before it gave a convulsive shiver in every limb and lay as rigid and lifeless as if it had been struck by lightning.”

Before we comment on the sudden death of the terrier, let us see what befell Enoch Drebber, who was forced by Jefferson Hope to swallow a pill containing curare. The wretched Drebber met the same fate as the aged terrier, for we find Watson writing: “... the action of the alkaloid is rapid. A spasm of pain contorted his features; he threw his hands out in front of him, staggered, and then with a hoarse cry, fell heavily upon the floor. I turned him over with my foot, and placed my hand upon his heart. There was no movement. He was dead.”

The deaths of Drebber and the terrier were dramatically portrayed, and Dr. Watson is to be congratulated on the vivid picture he presented. The trained scientist however, could not entirely accept the events as he outlined them. The difficulty lies in the fact that curare is relatively harmless if taken by mouth. If extremely large doses are administered on an empty stomach, sufficient curare may be absorbed to cause grave symptoms, but death would not be instantaneous, for absorption is slow from mucous surfaces. If, on the other hand, there were an open lesion in the stomach or the upper part of the small intestine, such as an ulcer, then rapid absorption could take place, and death would ensue in a relatively short time, although not as rapidly as portrayed in the story. The action of curare is rapidly lethal only if injected directly into the blood stream. It would stretch our credulity too far to assume that both the dog and the man had either a gastric or a duodenal ulcer.

Let us now consider how curare was employed in the story *The Adventure of the Sussex Vampire*. It will be recalled that Sherlock Holmes was asked by Mr. Robert Ferguson to investigate certain irregularities in his household. Ferguson, a fine gentleman, was very much in love with his beautiful Peruvian wife; but one day, to his infinite horror, he had actually seen her sucking blood from a wound on the neck of their year-old baby. She refused to make any explanation, and the husband and wife became estranged. There was another child in the family, an invalid boy of fifteen, Ferguson's son by a previous marriage.

Dr. Watson accompanied Sherlock Holmes to Ferguson's country home on a dreary autumn day. One of the rooms of the old house contained a fine collection of South American utensils and weapons which presumably had been brought from Peru by the mistress of the house. As Holmes was examining this interesting collection, the movements of a dog attracted his attention. Holmes noted that the dog experienced difficulty in walking. The astute detective asked Ferguson what ailed the dog. His host replied that the thing had also puzzled the veterinarian. The latter had thought it might be spinal meningitis. Holmes asked a few more questions about the dog, and finally remarked that the picture the dog presented was very suggestive.

The frantic husband insisted that Holmes tell all he knew or suspected. Holmes then gently explained to the indulgent father that his fifteen-year-old invalid boy was so insanely jealous of his healthy baby half-brother that

he had tried to do away with him by wounding him with an arrow treated with curare. The boy had first tried out the poison on the dog. Ferguson's wife, in an heroic effort to save her baby, had sucked the site of the arrow wound. The mystery was solved, and Holmes and Watson had the keen satisfaction of clearing up the grave misunderstanding between Ferguson and his lovely Peruvian wife.

In this story, curare was handled in an expert manner. It is known, of course, that the South American Indians dipped their arrowheads into a curare solution before using them to kill birds. The curare was rapidly absorbed from the wound made by the arrow; the wing muscles became paralyzed; and the bird plummeted to earth—an airplane without wings.

The Adventure of the Sussex Vampire was written many years ago. What is the status of curare today? This interesting compound has actually insinuated itself from the jungle not only into the experimental laboratory but into the surgical amphitheater as well. This is not the place to take up in detail the clinical use of curare, but a few remarks are in order.

Relaxation of the muscles is often highly desirable in surgical procedures; it is not surprising, therefore, that the anesthetists have added curare to their armamentarium. Although curare has no anesthetic action *per se*, it has been found to serve as a useful adjuvant to certain anesthetic agents. It has been used also for the convulsions of strychnine poisoning, tetanus, and hydrophobia, as well as certain spastic contractures. It could furthermore be of help in the management of dislocations, especially in heavily muscled individuals.

Curare should be employed only by experienced workers, for the muscles of respiration may become paralyzed, and unless mechanical respiration is given immediately, the patient will die of asphyxia. Fortunately, curare is quickly excreted by the body, and the patient will soon start voluntary breathing movements.

In conclusion, let us turn to the tales. *The Adventure of the Sussex Vampire* could have been written only by an individual quite familiar with the action of curare, and with a good understanding of clinical medicine. This story is of especial interest to physicians. Not only is curare used in the plot in an interesting and unusual manner, but certain psychosomatic problems are presented: The estrangement of the husband and wife; and a

clear portrayal of how the mind of a physically handicapped youngster may become warped, even though reared in an excellent environment.

SHERLOCK HOLMES AND THE PORTUGUESE MAN-OF-WAR

“I did what I could to relieve his pain.”

The Adventure of the Copper Beeches

The Adventure of the Lion's Mane has always appealed to me, although I am mindful that this story is not universally popular with Holmesian enthusiasts. The story was written by Sherlock Holmes himself. The title alone arrests one's attention. Immediately, one thinks of a magnificently maned lion at bay, or, as one artist pictures him, standing in the moonlight on the shore of the ocean, roaring defiantly at the incoming waves.

As the adventure unfolds, however, we learn that while the story does deal with the ocean, a lion is not mentioned, but rather a huge jellyfish, *Cyanea capillata*. It is also called the “Lion's Mane,” for as Holmes wrote, the jellyfish resembled a mass of tangled hair which looked as if it might have been procured from a lion's mane.

Let us recall that in this story the death of a person, as well as that of a dog, was produced by the poisonous sting of the Lion's Mane. The victim of this catastrophe was Fitzroy McPherson, a young scientist who had suffered from rheumatic fever, and whose heart had been damaged. In spite of his cardiac ailment, he was portrayed as a fine athlete and an expert swimmer. It was his custom to take a daily dip in the ocean in and out of season.

At the time of which we write, Sherlock Holmes was living in retirement on the Sussex Downs, not far from the place where McPherson often swam. One fine morning in July, following a severe storm, Holmes and his neighbor Stackhurst, who kept a preparatory school, were taking a bracing walk. To their intense surprise and dismay, they discovered the figure of McPherson, a science master of Stackhurst's establishment, coming up the path staggering as if drunk. As they came nearer, he uttered a terrible cry and fell upon his face. They rushed to his side and turned him

on his back. He was *in extremis*. Just before he expired, he indistinctly uttered words that sounded like “lion’s mane.” The cause of his death was a mystery even to Holmes.

A few days later, Ian Murdoch, a colleague and close friend of the late McPherson, went swimming in the exact spot where the latter had met his untimely death. Holmes dramatically describes how Ian Murdoch lurched in the room, extremely pale, and with his clothes all rumpled. It was an effort for him to stand; finally, he staggered to the sofa and collapsed from the excruciating pain. He experienced great difficulty in getting his breath, his face appeared livid, and beads of cold sweat stood on his brow. It appeared that he was also *in extremis*.

It is not necessary to relate here the steps which Holmes took in solving this baffling mystery. Suffice it to say that he came to the conclusion that both McPherson and Murdoch had come into contact with a huge jellyfish. He became absolutely convinced of this when he discovered in the attic of his home a book by J. G. Wood entitled, *Out of Doors*. Holmes found in this book a description of a patient who had been in contact with a jellyfish. The victim complained that while the local pain was severe, it was nothing compared to the pangs which surged through his chest. It was impossible for him to stand up. The cardiac beat became quite irregular; the heart would virtually stop, and then several violent pulsations would occur.

Admittedly, Holmes’ story is a thrilling one. The criticism to be made, however, is that although *Cyanea capillata* can produce a nasty sting, it probably could not produce death, especially in a healthy person. There is no question but that it would be an unpleasant experience for a swimmer to come into contact with *Cyanea*. In a standard textbook of biology, *Cyanea arctica*, another species of the same genus, is described as a creature which may measure six feet in diameter and bear tentacles reaching the astonishing length of one hundred and thirty feet! It can readily be seen that a swimmer could easily be stung by such a jellyfish before recognizing the danger.

Since it is generally believed that *Cyanea capillata* cannot cause the death of an individual, and since the symptoms described in *The Adventure of the Lion’s Mane* were so grave, the coelenterate which Holmes had in mind probably was no other than *Physalia*, commonly known as the

Portuguese man-of-war. Exposure to the tentacles of this creature is apt to produce a chain of alarming symptoms.

Owing to my warm interest in Holmes' story I read with profit and keen enjoyment a case history reported by Klein and Bradshaw.^[2] A twenty-year-old man, while swimming about twenty feet off North Miami Beach, Florida, came into contact with a Portuguese man-of-war. His attention was drawn to a purple object floating near him, which he thought to be a balloon. He swam up and touched it. Immediately, he became aware of a sharp stinging sensation on his arm and shoulder. Noting several strands which he could not brush off, he left the water at once, and a companion removed them with a towel. Underneath each strand could be seen a "painful fiery red welt." Soon he found it difficult to breathe, and shortly suffered from severe abdominal cramps. A little later, he showed signs of shock and mental confusion.

A swimmer exposed to the tentacles of these large jellyfish may, according to Klein and Bradshaw, suffer pain, swelling, and redness in the affected part. In a few minutes to an hour following exposure, systemic effects may appear, such as anxiety, muscular pains and cramps, dyspnea, constriction of the throat, cardiac symptoms, and prostration. The authors stress that the alarming symptoms which often follow stings from coelenterates suggest the action of a powerful neurotoxin.

Waite,^[3] writing in the same year, stresses the fact that contact with a Portuguese man-of-war produces symptoms indicative of anaphylactic shock, such as pallor, sweating, faintness, fall in blood pressure, and the like. He mentions that although there have been no authenticated medical reports of death as a result of the sting of a Portuguese man-of-war, it could happen if complicated by anaphylactic shock.

One wonders, of course, what would have happened if the patient described by Klein and Bradshaw had been some distance from shore. If anaphylactic shock had occurred within a few minutes, the patient no doubt would have drowned if no one had been in the vicinity to help him. A swimmer who experiences difficulty in breathing, who is afflicted with severe abdominal cramps, who shows signs of shock, and who is mentally confused, will certainly drown. Indeed, any one of these four symptoms could provoke disaster in deep water.

The treatment instituted by Klein and Bradshaw is worthy of our attention. The patient was hospitalized and given calcium gluconate and benadryl intravenously. Epinephrine and atropine were administered intramuscularly, and ammonia applied locally. I might add that a purist could raise the question whether all this medication was necessary. The next day, the itching areas were treated with tetracaine. The patient made an uneventful recovery, although it took about three weeks before the lesions were healed. Waite emphasizes that, in case of anaphylactic shock, epinephrine or benadryl should be given immediately. He warns further that if angioneurotic edema appears, the air passageways should be cleared, oxygen supplied, and, if necessary, a tracheotomy performed. The latter constitutes heroic treatment, but may, of course, save a person's life.

Let us return to *The Adventure of the Lion's Mane*, in an attempt to ascertain what medical agents were used fifty years ago to treat the sting of coelenterates. As previously mentioned, Sherlock Holmes was in retirement and living alone, so unfortunately he could not call upon his friend Dr. Watson to prescribe for Ian Murdoch.

Holmes, however, rose to the occasion and made an earnest attempt to help the patient, for he writes that he soaked cotton in salad oil and applied it to the wounds. He felt that this greatly alleviated the pain. The famous detective showed real ingenuity in making use of such a homely remedy. He also gave the suffering man liberal quantities of brandy, which doubtless had a narcotizing effect. There is considerable evidence that brandy was widely used for medicinal purposes at the turn of the century, when this adventure is supposed to have occurred.

It is of interest to speculate upon what medication Dr. Watson would have administered to Ian Murdoch had he been in attendance. At that time, neither benadryl nor calcium gluconate nor, for that matter, tetracaine was available. On the other hand, epinephrine, atropine, and ammonia were even then widely used. Dr. Watson presumably would have given a hypodermic injection of morphine and offered the patient brandy, as did Sherlock Holmes. He, too, doubtless would have applied some soothing ointment to the painful lesions, or, like Klein and Bradshaw, might have used ammonia. It is even possible that he would have employed epinephrine. The latter certainly was indicated, for the symptoms described by Holmes were those of anaphylactic shock. This syndrome (anaphylactic

shock) was not well understood fifty years ago. In point of fact, the word “anaphylaxis” was coined in 1907 by Richet. This, as mentioned earlier, was the year when the adventure was supposed to have taken place. Holmes, however, wrote *The Adventure of the Lion’s Mane* probably in 1925, since it was first published in 1926. At that time, the significance of anaphylactic shock in man was pretty well understood, and it is possible that he was familiar with this condition.

It is a pity that we have to resort to speculation, and that Dr. Watson was not in attendance instead of Holmes, a nonmedical man. If such had been the case, Watson probably would have mentioned the agents medical men used at that time. This would not only have satisfied our sympathetic curiosity but, what is more important, these facts would have been of historical interest. We then could actually have compared the medical agents which were prescribed fifty years ago with those currently employed for the grave symptoms of shock produced by the excruciatingly painful stings of coelenterates.

DOCTOR WATSON AND NERVOUS MALADIES

“... the shock of the night’s adventures had shattered his nerves.”

The Hound of the Baskervilles

The harrowing adventures experienced by some of the characters in the stories of Sherlock Holmes, and the great shocks they sustained, often induced in them a state of high nervous tension. References to such individuals are numerous.

Sherlock Holmes himself possessed an iron constitution, and was favored with an especially well-balanced mind. His biographer, John H. Watson, M.D., writes, “All emotions ... were abhorrent to his cold, precise but admirably balanced mind” (*A Scandal in Bohemia*). But even Holmes, on one occasion at least, was on the verge of a serious nervous breakdown. In the spring of 1897, Dr. Watson became concerned about the health of his distinguished friend: “... I found him a prey to the blackest depression ... [even his realization of his worldly fame] was insufficient to rouse him from his nervous prostration” (*The Reigate Puzzle*).

Watson felt that Holmes needed a complete rest in a quiet and soothing atmosphere, and finally persuaded the great detective to take a holiday in the country. Arrangements were made to stay at the home of one of Dr. Watson’s old friends. On the evening of their arrival, their host, Colonel Hayter, in the course of a conversation following dinner, told them of an unusual burglary which had occurred a few nights before in the neighborhood. When Holmes manifested too warm an interest in this event, Watson became alarmed and cautioned him: “You are here for a rest, my dear fellow. For Heaven’s sake don’t get started on a new problem when your nerves are all in shreds” (*The Reigate Puzzle*). Watson’s admonition proved of no avail. The next morning, while at breakfast, they were informed that a dastardly murder had been committed at the home of a nearby neighbor. Inspector Forrester, knowing that Sherlock Holmes was in the vicinity, asked for his help. Holmes answered the call, and after a rather

painful experience, solved the mystery brilliantly in the course of a few hours.

The neat solution of this case, together with the concomitant excitement which it afforded, apparently caused Holmes to recover quickly from his “blackest depression,” for that afternoon he remarked to his faithful friend, “Watson, I think our quiet rest in the country has been a distinct success, and I shall certainly return much invigorated to Baker Street tomorrow” (*The Reigate Puzzle*). This episode demonstrates the man’s superb constitution. Most people would have required several weeks, or even months, to recover their normal health following a severe nervous breakdown. But not Holmes.

In several instances, the characters described by Dr. Watson showed clear-cut signs of hysteria. When the well-known banker Alexander Holden found that the “Beryl Coronet,” which had been intrusted to his care, had been mutilated and three of its jewels stolen, he became greatly agitated and called on Sherlock Holmes:

For a while he could not get his words out, but swayed his body and plucked at his hair like one who has been driven to the extreme limits of his reason. Then suddenly springing to his feet, he beat his head against the wall with such force that we both rushed upon him and tore him away to the centre of the room.

The Adventure of the Beryl Coronet

This highly respected citizen must have been an emotionally unstable person. One wonders how he could have been a successful banker, with such a sensitive nervous system. The poor man must have lain awake many a weary night worrying about his investments. We do not know his subsequent history, but I suspect that he eventually developed hypertension or suffered from gastric ulcers.

When Rachel Howells was questioned by the master of the house about the disappearance of her perfidious lover Brunton, she showed pronounced signs of hysteria:

She looked at me with so strange an expression that I began to suspect that her brain was affected.... She fell back against the

wall with shriek after shriek of laughter, while I, horrified at this sudden hysterical attack, rushed to the bell to summon help. The girl was taken to her room still screaming and sobbing....

The Musgrave Ritual

There was a real reason for the girl's behavior; she had been directly responsible for her lover's horrible death. It is small wonder that she suffered a violent hysterical attack.

Once Dr. Watson was asked to see the wife of his boyhood friend Robert Ferguson. It was a tragic situation. She had been observed on several occasions sucking blood from her infant son. The sympathetic doctor stated that she appeared only half conscious, and had a frightened expression. Her pulse and temperature were both high, but he felt that this was due to her highly emotional state (*The Adventure of the Sussex Vampire*). Mrs. Ferguson was probably suffering from hysteria. The attack cleared up promptly when Sherlock Holmes sat down at the patient's bedside and, in the presence of her husband, analyzed the situation, explaining why it was necessary for Mrs. Ferguson to suck the baby's wounds. They had been made by arrows dipped in curare. The baby's insanely jealous half-brother had committed this atrocious deed. Truly, a shocking episode!

In *The Greek Interpreter*, reference is made to a man who showed symptoms of St. Vitus' dance. Mr. Melas described his unpleasant companion: "... his lips and eyelids were continually twitching like a man with St. Vitus' dance. I could not help thinking that his strange, catchy little laugh was also a symptom of some nervous malady."

The term "St. Vitus' dance" is seldom used nowadays; it is presently called "acute chorea." This condition is characterized by irregular involuntary contractions of the muscles and is associated with a variable amount of psychic disturbance. The name (St. Vitus' dance) has been handed down from the Middle Ages. Epidemics characterized by excitement, gesticulations, and dancing brought about mainly by religious fervor were in those days not uncommon. Whenever these symptoms became excessive, the people in the Rhenish province frequently made pilgrimages to the Chapel of St. Vitus in Zebern.

Another reference to twitching muscles may be cited. Sherlock Holmes called on Mr. Sidney Johnson, senior clerk and draughtsman, in an office from which extremely important secret papers had disappeared. Watson writes that the clerk had haggard cheeks, and that his hands were twitching from the nervous strain he had undergone (*The Adventure of the Bruce-Partington Plans*).

In the tales, other references may be found to shattered nerves and states of nervous exhaustion. In describing the condition of her father, Alice Turner tells Sherlock Holmes: "... Dr. Willow says that he is a wreck and that his nervous system is shattered" (*The Boscombe Valley Mystery*). The real reason for his pitiful state, which his daughter did not know, was that he had recently murdered an old acquaintance.

When Don Murillo, Tiger of San Pedro, tried to kidnap Miss Burnett, she broke away from him, and with the help of a friend got into a cab. Here Holmes saw her, and observed that she was in a state of collapse from nervous exhaustion (*The Adventure of Wisteria Lodge*). This brave lady had been locked in a room for a number of days with insufficient food, and had undergone other traumatic experiences. It is no wonder that Holmes found her in such a pitiful state.

After Sir Henry Baskerville had narrowly escaped death on the moor from the savage hound, he suffered a nervous collapse, "... the shock of the night's adventure had shattered his nerves." Sir Henry assuredly was not a weak man, but the victim of a concatenation of unfortunate circumstances. The family history of the hound, Sir Henry's love for a married woman, his isolation and loneliness, and the somber landscape of the bleak moor doubtless all preyed on his mind. When the hound actually sprang upon him, his nervous system finally reached the breaking point. Stronger men than he would have quailed.

Mr. Marlow Bates, the manager of the estates of the wealthy "Gold King" Mr. Gibson (whose wife had been found dead), called on Sherlock Holmes. Watson describes him as a small, thin, fidgety man, who appeared to be on the brink of a serious nervous breakdown. Although in great fear of his vicious employer, Bates detested him so heartily that he hurried to Baker Street and told Holmes that Gibson was an "infernal villain." In view of the circumstances, this was indeed a grave accusation. The reader will

recall that the “Gold King” had not murdered his wife; she had died by her own hand (*The Problem of Thor Bridge*).

The eyes often show a characteristic expression under emotional strain. Referring to Colonel Valentine Walther, whose brother had suddenly died, Watson observes that he had wild eyes and presented a worried and disheveled appearance (*The Adventure of the Bruce-Partington Plans*). It was discovered that it was the colonel who had stolen an extremely important state document. His older brother, Sir James, suspected Valentine’s guilt. The shock was too much for the older man and caused his death; there was a question as to whether he took his own life. Be that as it may, his younger brother was responsible for the tragedy. We are not surprised to learn that the eyes of the colonel had a wild expression.

People under great emotional strain often break out in a cold sweat. Nearly everyone has, at one time or another, experienced this phenomenon. It is caused by violent stimulation of the sympathetic nervous system. When John Openshaw’s uncle Elias received a letter containing five orange pips, a great change came over him. At times he acted like a madman. His nephew told Holmes, “At such times I have seen his face, even on a cold day, glisten with moisture, as though it were new raised from a basin” (*The Five Orange Pips*).

A cold sweat is more apt to manifest itself on the forehead, but it is not necessarily limited to the face, for any portion, or all, of the body may be involved. The cold, clammy sweat produced under conditions of intense mental stress is different, of course, from the normal healthy sweat brought about by a warm environment or by physical exertion, or by both. The physiologic explanation of a cold sweat is that it is a condition brought about by pain or fear as a reaction anticipatory of the strenuous muscle movements that may ensue.

Victor Trevor’s father was also the recipient of a letter which led to tragedy:

My father read it, clapped both hands to his head and began running round the room in little circles like a man who has been driven out of his senses.... I saw that he had a stroke.

The “Gloria Scott”

Apparently, this individual became so agitated, and his blood pressure rose so high, that he was stricken with a fatal cerebral hemorrhage. It is well known that emotions may produce a pronounced rise in blood pressure even in healthy people.

Most of the nervous maladies described by Dr. Watson were in the nature of acute episodes, and more or less transient in character. The individuals he depicted were high-strung, nervous people, overwhelmed by the traumatic experiences to which they had been subjected. Some of these were unable to suppress their emotions and either became hysterical or suffered a nervous collapse.

As a rule, we do not think of the English people as giving way to their feelings, but as governing them strictly. It is the Latins who are supposed to manifest hysterical reactions. However, it would take extraordinarily strong men, regardless of their nationality, to maintain normal mental equilibrium under some of the situations described in the spine-tingling tales; even the stolid Britishers cracked under the mental strain. The nervous system of man can withstand just so much and no more. It is given to but few to possess nerves of steel, and even such rare individuals finally reach a breaking point. Man is not a machine, but a human being.

If Dr. Watson were writing his adventurous stories today, he probably would stress the fact that long-continued nervous strain is likely to produce ulcers of the stomach or of the duodenum. The modern view is that mental worry may lead to hypertension and to diseases of the arteries, especially those which supply the heart muscles, namely, the coronaries—although it is true, as previously mentioned, that most of the characters depicted in the tales suffered mental strain for a substantial length of time.

It is noteworthy that both Sherlock Holmes and Dr. Watson maintained a nice mental balance even during periods of tremendous excitement and danger. A striking characteristic about both was their extraordinary fearlessness and mental poise. Although people around them often manifested signs of abject fear, or were given to hysterical outbursts, our heroes never faltered, but maintained a calm demeanor and dispatched neatly and efficiently the task before them.

DOGS AND SHERLOCK HOLMES

“It was a giant dog, as large as a calf, tawny tinted, with hanging jowl, black muzzle, and huge projecting bones.”

The Adventure of the Copper Beeches

To those interested in animals, it is gratifying to know that two of the most popular heroes of detective fiction—Sherlock Holmes and Dr. Watson—both loved dogs. They did not like them in a sickly, sentimental sense, but rather with a genuine, masculine affection. They regarded them as dogs, not as human beings, and emphasized and respected their canine personalities. I am sure every self-respecting dog would want to be so regarded.

Frequent mention of dogs is found in the tales. In point of fact, the grisly, spectral hound portrayed in *The Hound of the Baskervilles* has become a byword not only among readers of detective fiction but also with people in general. Various kinds of dogs are portrayed: large dogs, small dogs, good ones and bad ones. In one instance, a lovable, curly-haired spaniel is depicted; in another, a plodding bloodhound; and in still another, a fiendish dog.

One Sunday evening, in early September, in the year 1903, Sherlock Holmes summarily sent for Dr. Watson. The friendly doctor, not wishing to disappoint his friend, set out for Baker Street, and entered the old apartment, which for a number of years had also been his home. Holmes waved him to a chair, and after a considerable period of silence remarked that he was seriously considering writing a monograph on the use of dogs in the work of a detective (*The Adventure of the Creeping Man*).

Dr. Watson did not think much of the idea, and pointed out that this field had been pretty thoroughly explored. Holmes would not agree. He explained that he wished to approach the subject from a somewhat different angle than had previously been done. He argued that a dog reflected a family's life. If the household were a happy one, the dog would be friendly and frisky; conversely, if the family were gloomy, the dog would appear

sad. He insisted further that dangerous and unpleasant dogs are owned by dangerous and unpleasant people.

Holmes bolstered his argument by giving as an illustration the case on which he was presently working. He believed that the changed behavior of a wolfhound toward his master might turn out to be an important clue in solving the mystery. This eventually proved to be true, as will be described later. Although we may congratulate Holmes on his keen analysis, and on his pungent manner of presenting the argument, there surely must be many exceptions which come to the minds of us all. We have witnessed nice people who harbor vicious dogs, and the opposite as well. We will not quibble, but agree for the main part that Holmes was probably right.

Hounds are frequently mentioned. In *The Sign of the Four*, Watson, describing Holmes in action, writes: "So swift, silent, and furtive were his movements, like those of a trained bloodhound picking out a scent." Again, in *A Study in Scarlet*, we see: "As I watched him I was irresistibly reminded of a pure-blooded, well-trained foxhound, as it dashes backward and forward through the covert, whining in its eagerness, until it comes across the lost scent." In *The Red-Haired League*, Jones of Scotland Yard, speaking to his colleague Wilson, comments: "Our friend Holmes here is a wonderful man for starting a chase. All he wants is an old dog to help him in the running down." Other habits of hounds may be found in *The Adventure of the Devil's Foot* and *The Adventure of the Bruce-Partington Plans*. Numerically, hounds are mentioned more frequently in the tales than any other kind of dog, and in several instances hounds actually track down the criminal.

It will be recalled that in *The Sign of the Four* the dependable old hound Tobey follows a creosote scent. In *The Adventure of the Missing Three-Quarter*, faithful old Pompey follows a carriage which had had aniseed squirted upon the wheel by Holmes. On this occasion, Holmes remarks to Watson that a draghound would follow aniseed indefinitely. Holmes takes almost a human interest in Pompey. The detective formally introduces the dog to Dr. Watson and speaks highly of the draghound's tracking ability. However, he did not trust all dogs, for we find him saying to Watson that some dogs bite the hand that feeds them (*The Adventure of the Three Gables*).

Two especially fiendish dogs are portrayed in the tales: Carlo the mastiff, in *The Adventure of the Copper Beeches*; and the spectral hound, in *The Hound of the Baskervilles*. In the former, the despicable Rucastle, the master of Carlo, said: “We feed him once a day, and not too much then, so that he is always as keen as mustard.... God help the trespasser whom he lays his fangs upon!” Miss Violet Hunter, the heroine of the story, saw this beast one night: “It was a giant dog, as large as a calf, tawny tinted, with hanging jowl, black muzzle, and huge projecting bones.” Watson vividly describes the attack Carlo finally made on his master:

There was the huge famished brute, its black muzzle buried in Rucastle’s throat, while he writhed and screamed upon the ground. Running up, I blew its brains out, and it fell over with its keen white teeth still meeting in the creases of his neck.

In *The Hound of the Baskervilles*, we find a hound which would frighten the Evil One himself. Evidently, this huge beast was a mongrel: “It was not a pure bloodhound and it was not a pure mastiff; but it appeared to be a combination of the two—gaunt, savage, and as large as a small lioness.” Watson paints this animated picture:

A hound it was, an enormous coal-black hound, but not such a hound as mortal eyes have ever seen. Fire burst from its open mouth, its eyes glowed with a smouldering glare, its muzzle and hackles and dewlap were outlined in flickering flame.

While this hound was worrying the throat of Sir Henry Baskerville, Holmes emptied five barrels of his revolver into the creature’s flank. “With a last howl of agony and a vicious snap in the air, it rolled upon its back, four feet pawing furiously, and then fell limp upon its side.”

Other breeds are not ignored. Holmes, who had considerable professional jealousy, would not admit that Jones of Scotland Yard had the imagination a detective should have, or that he had a keen intellect, but does pay him this fine compliment: “He is as brave as a bulldog....” (*The Red-Haired League*.) A reference is made to rat terriers in *The Adventure of the Crooked Man*. Spaniels also are mentioned on at least two occasions. Dr. Mortimer, in the story of *The Hound of the Baskervilles*, owns a curly-haired spaniel of which he is extremely fond. Unfortunately, this lovable dog comes to grief on the moor. In *The Adventure of Shoscombe Old Place*,

a spaniel plays an important part in solving the mystery. The so-called “Shoscombe spaniels” were famous throughout England, and, according to Holmes, were frequently mentioned at dog shows. He successfully uses one of these, a pet spaniel, to establish the fact that the person in the carriage is not the dog’s mistress. Holmes was satisfied that the spaniel was right, and insisted that dogs do not make mistakes in such cases.

Another instance can be cited where the behavior of a dog helped Sherlock Holmes solve a perplexing problem. In talking to Watson, he laid great stress on the fact that Professor Presbury’s heretofore friendly wolfhound Roy had turned on his master and tried to attack him. This extraordinary episode played an important part in the final solution of the mystery (*The Adventure of the Creeping Man*). The facts in the case were these: Professor Presbury, a distinguished physiologist, had fallen madly in love with a very young girl, and, in an effort to regain his lost youth, had periodically been taking hypodermic injections of serum obtained from langurs. According to the story, the monkey serum produced apelike qualities in Professor Presbury. Following the injections, the professor would deliberately and maliciously tease his wolfhound until one night the hound slipped his collar and viciously attacked his master. He buried his muzzle in his master’s throat; and if timely aid had not come, Presbury would have been killed. As Holmes remarked to Watson, the hound thought he was attacking the monkey and not the professor.

Since Holmes was a detective, and not a professional biologist, he seldom found it necessary to use dogs for experimental purposes. Once, however, he did not hesitate to try out the effects of a poison drug, curare, on an old dog which was *in extremis*. It will be recalled that this poison caused instant death (*A Study in Scarlet*). While not a biological scientist—as we understand the word today—Holmes nevertheless was scientifically minded, and it is certain that had the need arisen to obtain evidence which would have helped mankind he would not have hesitated to experiment on a normal, healthy dog. He was a practical man, in the fullest sense of the word, and had, as Watson said, “... an admirably balanced mind” (*A Scandal in Bohemia*).

It is true that both Sherlock Holmes and Dr. Watson grieved when Dr. Mortimer’s pet curly-haired spaniel met its fate on the moor, and they both loved Tobey, the faithful old bloodhound who had aided them in *The Sign of*

the Four. However, they did not hesitate to shoot five bullets into the flank of the hound of the Baskervilles when he was at the throat of Sir Henry, or to blow out the brains of Carlo the mastiff when he sank his teeth into the fat neck of the villainous Rucastle. Holmes and Watson were not sentimentalists, but virile, vigorous men of action—the type of men dogs like.

Finally, Dr. Watson pays a tribute to dogs in general in *The Adventure of the Lion's Mane*. In this story, after McPherson died from his encounter with the Lion's Mane (*Cyanea capillata*), his pet Airedale eventually met a similar fate. According to Sherlock Holmes himself, the terrier did not eat for a week following his master's death, and finally followed the trail of his dead master. He was found dead on the edge of the same pool where his master had lost his life. It is true that the terrier did not have the opportunity of saving his master's life—but no doubt he would have, had it been possible. The story portrays vividly the bond of companionship between a man and his dog, for in this instance the dog apparently gave his life searching for his master. Dr. Watson pays tribute to such devotion in the words of Sherlock Holmes: "That the dog should die was after the beautiful, faithful nature of dogs." This is a tribute that can be endorsed by all, but especially by biologists, who are so deeply indebted to the dog for the aid it has given in the search for means to alleviate the pain and suffering of mankind. I am certain that every biologist who professes to be a Christian gentleman would say "Amen" to Dr. Watson's beautiful tribute to an animal which would die so that mankind might live.

THE BOTANICAL DOCTOR WATSON

“... there stood a patriarch among oaks, one of the most magnificent trees that I have ever seen.”

The Musgrave Ritual

Many delightful references to plant life may be found in the tales. These allusions enrich the stories and add greatly to their charm and interest. Love for inviting landscapes, for trees, for flowers, and for shrubs appears to be innate in human beings. This would be a drab world indeed without a variegated and a beautiful plant life. One need not be a botanist to appreciate the beauty and fragrance of a rose or a violet. They are different in design, but lovely in their own ways. It is not meet that we attempt to analyze their beauty; let us merely enjoy it. If an extravagant phrase be permitted, let us not subject the beauty of the orchid to the flame of the analytic blowpipe or attempt to measure the fragrance of the June rose with an olfactometer.

After Dr. Watson had lived with Sherlock Holmes a few weeks, he became intensely interested in this quiet but unusual man. One day he sat down and made an attempt to analyze and evaluate Holmes' conversance in several fields. While Dr. Watson felt that Holmes' understanding of anatomy was good, his knowledge of botany was “variable.” The worthy doctor writes: “Knowledge of Botany—Variable. Well up in belladonna, opium, and poisons generally. Knows nothing of practical gardening” (*A Study in Scarlet*). Thus, it appears that Holmes was more interested in medicinal plants, and especially those from which poisons could be extracted. This does not surprise us, because of his interest in crime detection; we know, too, of his researches in the chemistry of the alkaloids.

We must keep in mind that Watson wrote of Sherlock Holmes: “Appreciation of nature found no place among his many gifts” (*The Adventure of the Cardboard Box*). A man uninterested in nature is not apt to take especial notice or appreciate the beauty of flowers or vegetation. We

are indebted, therefore, largely to Dr. Watson for the many pleasing references to plant life and landscapes.

We find allusions to a botanist and to the study of botany in *The Valley of Fear* and in *The Adventure of Wisteria Lodge*. In the first story, Watson likens Holmes to a botanist rapt in his admiration for a beautiful flower; and in the latter story, Watson points out that many instructive days can be spent in the study of botany and suggests the use of an elementary text on the subject, a spud, and a proper box to hold the specimens. The amateur botanist will surely appreciate these suggestions. Such a person will find pleasure no matter where he strays—in the forest, the plains, the mountains, or the desert. He will never be bored.

Dr. Watson obviously loved the vegetation of early spring, for he writes: “... the first faint shoots of green were breaking out upon the elm, and the sticky spearheads of the chestnuts were just beginning to burst into their fivefold leaves” (*The Adventure of the Yellow Face*). Also, we read, “The trees and wayside hedges were just throwing out their first green shoots.” (*The Adventure of the Speckled Band*).

The observant doctor was not unmindful of the autumn foliage. In *The Hound of the Baskervilles*, he calls our attention to the melancholy aspect of the countryside in the fall of the year, and speaks of the yellow leaves lying on the ground, some still fluttering down from the trees. In *The Problem of Thor Bridge*, he notes that the plane tree in the backyard was losing its remaining leaves on a stormy October morning. (We would call a plane tree a sycamore.) Some people claim they become somewhat melancholic in the autumn and dislike to see the leaves falling, the grass turning brown, and other signs associated with the fall of the year. I am not one of these. There is much to be said for distinct changes in seasons. The changes occurring in the autumn are necessary so that the wonderful season of spring may be thoroughly appreciated and enjoyed.

Several types of landscapes are pictured. In *A Study in Scarlet*, the arid land in our great West is described:

As far as the eye can reach stretches the great flat plainland, all dusted over with patches of alkali, and intersected by clumps of the dwarfish chaparral bushes.

The purist might object to the term “chaparral bushes,” because chaparral, being a nonspecific term, refers to any stunted type of vegetation found growing on dry soil in our West. It might have been more proper simply to have said “clumps of chaparral.” In the same story, a prosperous, peaceful countryside landscape is depicted: “All looked so peaceful and happy, the rustling trees and the broad silent stretch of grainland....” In *The Adventure of the Solitary Cyclist*, a landscape of flowering gorse is described, and attention is called to its golden color on the heather-covered fields.

Dr. Watson obviously was impressed with landscapes of gold and bronze, especially those which contained faded ferns, for we find him using the description several times, for example:

The sun was beginning to sink ... and the long, sloping plain in front of us was tinged with gold, deepening into rich, ruddy browns where the faded ferns and brambles caught the evening light.

Silver Blaze

Similarly, in the same story: “... the low curves of the moor, bronze-coloured from the fading ferns, stretched away to the sky line....” Besides these landscapes, several references to some of the famous forests of England are made in *The Adventure of Black Peter* and *The Naval Treaty*.

As might be anticipated, numerous allusions are made to trees. Dr. Watson presents a pleasing picture: “Just beyond it is a nice little grove of Scotch firs, and I used to be very fond of strolling down there, for trees are always a neighborly kind of thing” (*The Yellow Face*). The word “neighborly” strikes a sympathetic note. All of us have experienced this feeling at one time or another. Walking in a pleasant woodland, one can quietly commune with nature. The singular thing is that on such occasions the creative mind is often stimulated. New ideas and new concepts apparently rise from the subconscious to the higher brain centers. Numerous examples could be given of this truly remarkable phenomenon.

Dr. Watson must have had a lively interest in oak trees, for he alludes to them on several occasions. He speaks of “ancient” oaks and “mighty” oaks,

terms surely applicable to the famous English oak trees (*The Valley of Fear* and *The Adventure of the Solitary Cyclist*).

The prize oak, presumably, is the one described in *The Musgrave Ritual*. Sherlock Holmes related the story to Dr. Watson one winter night: "... there stood a patriarch among oaks, one of the most magnificent trees that I have ever seen." He went on to quote his host, Reginald Musgrave: "It was there at the Norman Conquest in all probability.... It has a girth of twenty-three feet." Truly, it was a mighty oak and worthy of comment. If its girth was twenty-three feet, it had a diameter of approximately seven and one-third feet. The events described in the story occurred in the last quarter of the nineteenth century, so the tree must have been over nine hundred years old if it dated from the Norman Conquest of 1066.

Probably few people appreciate the longevity of oak trees. I recall, as a child, an oak that stood not far from my home. I have seen it periodically for over a half-century. It still looks about the same as when I first saw it, but presumably it has had a slow but steady growth throughout the years. Assuming that such a tree is not ravaged by wind and storm or struck by lightning, it probably will stand for several centuries. This is a fascinating thought, because it staggers the imagination to think what historical events could occur during such a span of years.

Dr. Watson describes trees in various settings, such as parks (*The Adventure of the Solitary Cyclist* and *The Adventure of the Abbey Grange*), and in several instances as borders for avenues or lanes. A number of examples might be cited, but one will suffice: "The house ... with a fine lime-lined avenue leading to it" (*The "Gloria Scott"*).

In *The Hound of the Baskervilles*, the Yew Alley of Baskerville Hall played an extremely important part in the story. It will be recalled that Sir Charles, before going to bed at night, was wont to take a walk down the Yew Alley. One night he saw the huge hound which the evil Stapleton had purposely let loose on the moor. Sir Charles became so frightened that he ran headlong and dropped dead from a heart attack.

In this story, too, mention is made of trees which have been subjected to adverse growing conditions. Dr. Watson speaks of trees which have been stunted and nipped on the moor, and emphasizes how the oaks and firs are bent and twisted by the furious storms throughout the years. The bizarre

shapes which such trees assume are truly fascinating. Their irregular outlines appear ghostlike by night and weird by day. They are the joy of the amateur photographer and the professional artist.

Various types of trees are used to create a proper setting for the mystery stories. In *The Adventure of the Mazarin Stone*, a melancholy grove of half-grown pines is described; and in *The Adventure of Wisteria Lodge*, a row of chestnuts is depicted as giving the avenue a gloomy appearance.

The fact that pine trees are used to set the note for a gloomy landscape calls for comment. I have always felt that a clump of evergreen trees around a house creates a distinctly depressing atmosphere; this is especially true if the house is isolated in the country. To me these somber trees give a sense of loneliness and gloom, and bring to mind Anna Katherine Green's mystery story, *The House of Whispering Pines*. It may be that this story, which I read in my youth, is the reason for my feeling. I am sure, in any event, that many people enjoy evergreen trees around their home, especially in winter, as a welcome relief from the drab, slate-gray landscape.

Besides those trees already enumerated, several other species are mentioned throughout the tales: namely, the larch, the beech, and the copper beech. In *The Adventure of the Copper Beeches*, this description occurs: "The group of trees with their dark leaves shining like burnished metal in the light of the setting sun...."

Flower beds and borders are described in several of the tales. In *The Sign of the Four*, we find that: "... just under the window a single foot-mark was visible in the flower bed." Also, in *The Adventure of the Devil's Foot*, a flower border lying underneath the window is noted, but it contained no trace of a footprint. Obviously, footprints are a necessary component of mystery stories; and what better place to look for them than in the soft earth of a flower bed under a window?

In all the stories, only a few flowers are specifically named: crocuses, orchids, roses, and violets. In *The Hound of the Baskervilles*, when Dr. Watson met the wife of the naturalist Stapleton on the moor, she asked him to pick an orchid for her, and explained that the moor was rich in them. This is supposed to have occurred in the fall of the year. It seems singular that orchids would be in bloom at that time.

The amateur gardener will be sympathetically interested to read of gardens and lawns which did not live up to expectations: “A small garden sprinkled over with a scattered eruption of sickly plants separated each of these houses” (*A Study in Scarlet*). In *The Adventure of the Retired Colourman*, a sorry-looking garden is depicted. It is pictured as going to seed and bearing every evidence of gross neglect. In *The Red-Haired League*, an ill-kept lawn is described: “... where a lawn of weedy grass and a few clumps of faded laurel bushes made a hard fight against a smoke-laden and uncongenial atmosphere.”

Several welcome allusions are made to the odors of flowers or of vegetation. The scent of flowers, gardens, greenhouses, trees, meadows, swamps, and the decay of autumn leaves often bring back memories of other times and places. In *The Naval Treaty*, we find “... the rich scent of the garden ...”; and in *A Study in Scarlet*, we read of “... the balsamic odours of the pine trees...” All of us have experienced the refreshing and delightful odors of a flower garden, and those who have been in the north, or in the mountains in almost any latitude, remember with pleasure the clean smell of the evergreen trees. In *The Adventure of the Lion’s Mane*, our attention is called to the scent of thyme on the downs. I am sure that to many people the faintly exotic aroma of thyme (which belongs to the mint family) is enjoyable.

Some plants have a distinctly heavy, cloying odor—for example, certain lilies. The effect produced may actually be unpleasant. Dr. Watson, in describing a greenhouse, brings this out (*The Adventure of Charles Augustus Milverton*). He speaks of the heavy fragrance of certain plants which actually causes a choking sensation. The odor of decayed plants arising from the treacherous Grimpen Mire is emphasized in *The Hound of the Baskervilles*. Most of us are familiar with the miasmatic vapors which often arise from swampland. Presumably, everyone has his own favorite odor. Some people enjoy the exquisite fragrance of the rose, some that of the carnation, and others the faint and delicate odor of the sweet pea. The city dweller who was raised in the country may yearn for the sweet smell of newly mown hay, and, strangely enough, may even long for the pungent ammoniacal smell of the barnyard.

Throughout the tales, we find mention of bracken, bramble, bushes, ferns, gorse, heather, hedges, laurel, lichens, rhododendron, shrubs, and

vines. We have already alluded to many of these. To my knowledge, in only one instance is there a reference to blood-stained vegetation indicating that a murder might have been committed. We find, in *The Adventure of the Priory School*, that Holmes, to the horror of Watson, held up some flowering gorse which was blood-stained. The surrounding heather, too, showed evidence of old blood stains. It will be remembered that these were found on the spot where the unfortunate German master of the Priory School had met his death.

Everyone interested in Sherlock Holmes will vividly recall the meeting between Holmes and Professor Moriarty on the brink of the falls of the Reichenbach. Dr. Watson inspected the spot shortly after the meeting:

A few yards from the end the soil was all ploughed up into a patch of mud, and the brambles and ferns which fringed the chasm were torn and bedraggled.

The Final Problem

The patches of mud and torn vegetation gave clear evidence of what had taken place on the edge of the falls. We learn later, of course—much to our surprise and satisfaction—that Holmes had, by his superior skill, triumphed over his arch enemy. The meeting marked the end of Professor Moriarty, the Napoleon of crime.

Dr. Watson's many allusions to botany give added charm to the tales, and indicate not only his love for the flora of England but also his zest for nature.

THE SURGICAL DOCTOR WATSON

“... the thin white hand he laid on the mantelpiece ... was that of an artist rather than of a surgeon.”

The Resident Patient

Dr. Watson professed to be a general practitioner of medicine. Such a person is called upon from time to time to do minor, but under ordinary conditions he does not attempt to do major, surgery. To my knowledge, there is no particular reference to major surgery in any of the tales, although one or two famous surgeons are mentioned. On the other hand, numerous allusions are made to minor surgery. I am cognizant of the fact that there is often but a thin line between major and minor surgery.

Dr. Watson evidently felt that a good surgeon must have large and supple hands, for he writes: “... the thin white hand he laid on the mantelpiece ... was that of an artist rather than a surgeon” (*The Resident Patient*). The size of the hand probably is not of major importance. It is the ability to use the hands that counts. Several eminent surgeons, to my own knowledge, are men of large physical stature, and I am sure have correspondingly large hands. But I have seen some equally capable men, hardly of average height, who had small hands. Harvey Cushing, the brilliant Harvard brain surgeon, was a man of medium size, and, if I recall correctly, had rather small hands. Let us not labor this point further, for this is not an essay on surgeons’ hands, but turn to the tales of Sherlock Holmes.

It is axiomatic that every man who practices medicine, regardless of his specialty, must be familiar with the current literature in his field. It is gratifying that Dr. Watson appreciated this. In *The Adventure of the Golden Pince-Nez*, Watson observes that one stormy night he and Holmes sat together in silence; the latter was working on a palimpsest, while Watson was reading a treatise on surgery.

Holmes appreciated that a correct diagnosis is of paramount importance to a conscientious surgeon. In *The Problem of Thor Bridge*, we find Mr. Neil Gibson, the “Gold King,” telling Holmes that he was like a surgeon,

because Holmes wanted to know every symptom before he ventured a diagnosis. Holmes quickly retorted that this was quite true, and strongly implied that his client was withholding important information which Holmes needed before the mystery about which he was being consulted could be solved.

There are a number of references to scalp wounds or injuries to the skull, or both. In the rough-and-tumble life a detective leads, it might be expected that he would often see such injuries. Let us examine some of them.

Once Sherlock Holmes himself came under a surgeon's care. The great detective had been attacked on the street by two ruffians and had been pretty roughly handled (*The Adventure of the Illustrious Client*). Sir Leslie Oakshot, the eminent London surgeon who had been called in, stated that Holmes had suffered lacerations of the scalp and that several stitches were necessary. Holmes fortunately made an uneventful recovery, and we learn that the stitches were taken out on the seventh day.

Dr. Watson describes a man in *The Adventure of the Solitary Cyclist* who had suffered a severe scalp wound. He had been found unconscious, although the bones of the skull had not been penetrated. Another case is mentioned which surely must puzzle the layman (*The Adventure of the Dancing Men*). A bullet passed through the frontal portion of the brain. The reader is assured that, although the patient was unconscious, she would live. The medically trained person would think immediately of an accidental lobotomy.

One is reminded in this connection of the skull of a workman exhibited in the museum of the Harvard Medical School. This rare specimen shows an enormous hole made by a tamping bar. The bar passed through the skull, and a considerable amount of brain substance was destroyed. Remarkably enough, the patient did not die from this terrific injury, but lived for a long time afterward—surely a most unusual case. It should be added that he suffered an unfortunate change in his personality following the accident.

In several instances, the head injuries mentioned were immediately fatal. For example, in *The Adventure of the Empty House*, an expanding revolver bullet was fired into the victim's head. In *The Boscombe Valley Mystery*, we find that: "The head had been beaten in by repeated blows of

some heavy and blunt weapon.” In this last story, Dr. Watson gets technical: “In the surgeon’s deposition it was stated that the posterior third of the left parietal bone and the left half of the occipital bone had been shattered by a heavy blow from a blunt weapon.” But contrast the following nontechnical description: “The injury from which the unfortunate veteran was suffering was found to be a jagged cut some two inches long at the back part of his head, which had evidently been caused by a violent blow from a blunt weapon” (*The Crooked Man*). This is a good example of a wound so simply described that anyone could easily understand and appreciate it. With few exceptions, Watson never forgot that he was writing for laymen.

Scalp wounds and injuries to the head are often used to dramatize detective stories. The average layman instinctively feels that such wounds are extraordinarily dangerous, and that people experiencing such accidents are nearly always in a critical state. This, of course, may or may not be true, because nature has provided a strong bony cage to protect the brain.

In *The Adventure of the Sussex Vampire*, there occurs a reference concerning an orthopedic problem. Dr. Watson writes that the gait of a fifteen-year-old boy suggested plainly that he had a weak spine. We are further informed that a fall in childhood had brought about this condition. The devoted father spoke of the lad as being an unfortunate cripple and quite inoffensive. He was quite mistaken, for this so-called inoffensive lad became so insanely jealous of his normal, healthy baby half-brother that he tried to do away with him by use of curare, the South American Indian arrow poison. When Holmes unearthed this diabolic scheme, he reluctantly informed the boy’s unsuspecting father. Holmes prefaced his remarks by stating that he was a busy man and would come to the point quickly; and he used the simile that, the swifter surgery is done, the less painful it is.

The idea Holmes expressed in regard to swift surgery holds our interest. In the early days, before the discovery of ether, surgeons for obvious reasons attempted to perform operations in the shortest possible time. Indeed, the skill of a surgeon was largely gauged by the speed with which he could perform an operation.

In at least two of the tales, the carotid artery was the focal point of interest. In *The Adventure of the Creeping Man*, we find that when Professor Presbury was viciously attacked by his wolfhound, the teeth of

the dog narrowly missed the carotid artery. In spite of the rather severe hemorrhage which followed, it will be recalled that the professor recovered. In *The Adventure of the Golden Pince-Nez*, the victim was not as lucky. His carotid had been severed by an old-fashioned sealing-wax knife, and death ensued in a short time. Watson emphasized that the wound was small but deep. He was aware that a large blunt instrument might have pushed the elastic artery aside rather than piercing it.

We find one reference to an amputation, although it was not performed by a surgeon. Early one morning, there appeared in Dr. Watson's consultation room a young engineer who, during the preceding night, had undergone a harrowing experience in which his thumb had been hacked off by a vicious counterfeiter. The doctor was astounded when he saw the wound: "It gave even my hardened nerves a shudder to look at it. There were four protruding fingers and a horrid, red, spongy surface where the thumb should have been. It had been hacked or torn right from the roots" (*The Adventure of the Engineer's Thumb*).

When Dr. Watson asked the victim whether the wound had bled a great deal, the answer was: "Yes, it did.... I tied one end of my handkerchief very tightly around the wrist and braced it up with a twig." Watson's succinct reply was: "Excellent. You should have been a surgeon." To satisfy the reader's sympathetic curiosity, it should be mentioned that Dr. Watson gave the patient a drink of brandy, and cleansed and bandaged the wound. Since the thumb had been "torn right out from the roots," it seems that Dr. Watson should have taken a few stitches to bring the cut tissues together. No such mention is made, and it is to be hoped that the wound healed satisfactorily.

It is noteworthy that this engineer must have had a rugged constitution. Dr. Watson, after caring for the wound, took him to 221B Baker Street, where he joined Holmes and Watson in eating a substantial breakfast. Then, for the benefit of the consulting detective, he related the night's terrible adventure at some length. He next accompanied Holmes and his companions on a train journey to Eyeford, a little Berkshire village. This was a strenuous program. The question arises, why did the kind Watson allow his patient to do all these things? This young man should have received a narcotic to deaden the pain, which must have been severe, and then should have gone to bed. If the patient experienced difficulty in going

to sleep, a hypnotic should have been administered. In this instance, Dr. Watson placed himself in an indefensible position.

Let us now turn from injuries of the human flesh and consider something relatively less gruesome. A surgical instrument is described and, interestingly enough, the name of the manufacturer is given. Holmes himself calls our attention to it: "... an ivory-handled knife with a very delicate, inflexible blade marked Weiss & Co., London" (*Silver Blaze*). Parenthetically, it should be stated that Weiss & Co., are still famous makers of knives. Indeed, there are several in my laboratory at the present time.

When Holmes speaks of the singularity of this knife, and suggests to Dr. Watson that surely this was in his line, the latter states, "It is what we call a cataract knife." Holmes replies, "A very delicate blade devised for very delicate work." The reader should be reminded that the knife in question was not employed to murder a human being, but a villainous horse trainer had intended to use it to cut the tendon of a famous race horse. It will be remembered that the nefarious scheme was never carried out, for the highly nervous thoroughbred animal killed his trainer by kicking him on the head before the operation could be performed.

As far as I am aware, Watson did not make any specific mention of any other surgical instruments in the tales. He does, however, on one occasion make an allusion to a surgeon's choice of his instrument during an operation. He is commenting on Holmes' selection of tools when attempting to open the safe of the notorious blackmailer, Charles Augustus Milverton, and points out that Holmes chose his instruments with the same scientific precision as a surgeon about to perform an important operation (*The Adventure of Charles Augustus Milverton*).

In the present essay, we have discussed only those cases which are concerned with surgery. But in England the medical man is often spoken of as a "surgeon," rather than a physician or a doctor or a general practitioner. We find such expressions in the Sherlock Holmes stories: "And here comes the country surgeon ..." (*The Adventure of the Copper Beeches*). Numerous other references could be cited, but this one will suffice. Furthermore, a doctor's treatment room is spoken of as his "surgery." This latter expression is often found, especially in English novels. The villain Stapleton, in *The Hound of the Baskervilles*, tells Dr. Watson that he had been visiting with

Dr. Mortimer, and that while they were in his “surgery,” Dr. Mortimer had pointed out Dr. Watson through the window. In another instance, we find, “The surgery, where I used to make up my drugs ...” (*The Sign of the Four*). In this country, we would simply call it “the doctor’s office,” or if we wanted to be more specific, his “treatment” or “examining room.” These are minor differences, it is true, but they are of passing interest.

Some of the wounds so dramatically described by Dr. Watson surely became severely infected, and the patient’s suffering must have been considerable. Pasteur and Lister had already published the results of their brilliant researches, and the medical profession was well aware of the dangers of infections at the time. In spite of this, not much could be done once the wounds began to suppurate, for the antibiotics had as yet not been discovered. We may rest assured, however, that Dr. Watson did all he could to alleviate the patient’s pain and to promote healing of the wound.

SHERLOCK HOLMES, THE CHEMIST

“A formidable array of bottles and test tubes, with the pungent cleanly smell of hydrochloric acid, told me that he had spent his day in the chemical work which was so dear to him.”

The Red-Haired League

Mention is made from time to time in the tales of certain chemical problems which occupy the attention of the great detective. Moreover, the chemical desk and chemical paraphernalia in the rooms at 221B Baker Street are referred to frequently. Indeed, the reader would be disappointed if they were not mentioned, for he has come to accept them as he has the Persian slipper and the gasogene. Since we are living in an age of chemistry, it is pertinent and timely to review Sherlock Holmes' interest in this important field. It is, moreover, of certain intellectual and historical interest to examine—even though rather superficially—the status of any phase of chemistry during the last quarter of the past century, or, for that matter, of any period. At the time of which we write, biological chemistry was still in its infancy, but organic chemistry was already an important subject.

In *A Study in Scarlet*, it will be remembered that when Stamford took Dr. Watson to the laboratory of the medical school to meet Sherlock Holmes, they found him working on a chemical problem. So enthusiastic was Holmes that, even before Stamford had had an opportunity to present Watson, Holmes sprang to his feet and cried: “I’ve found it! I’ve found it!... I have found a reagent which is precipitated by hemoglobin, and by nothing else.” He then acknowledged the introduction, and made the famous remark, “You have been in Afghanistan, I perceive.”

Holmes proceeded next to give a demonstration of his new test:

“Now, I add this small quantity of blood to a litre of water. You perceive that the resulting mixture has the appearance of water. The proportion of blood cannot be more than one in a

million. I have no doubt, however, that we shall be able to obtain the characteristic reaction.”

As he spoke, he threw into the vessel a few white crystals, and then added some drops of a transparent fluid. In an instant, the contents assumed a dull mahogany color, and a brownish dust was precipitated to the bottom of the glass jar.

Holmes was so well pleased that he added, “Now we have the Sherlock Holmes’ test....”

At this point a sour note can be injected. Assuming that Holmes did discover a sensitive test for hemoglobin, it was nevertheless not a specific one for human blood. The blood of many animals contains hemoglobin. In a case of murder, for example, if blood were found on a cudgel, a knife, or on the clothing of a suspect, it would still have to be proved that it was human blood. This, incidentally, can now be scientifically shown, but it is a long, delicate procedure.

After Holmes had demonstrated his brilliant experiment, the trio sat down, and Holmes and Watson discussed the possibility of sharing rooms. Holmes volunteered, “I generally have chemicals about, and occasionally do experiments.” Parenthetically, it may be stated that Watson on later occasions complained about the odors produced by these experiments. As we know, however, an agreement was reached, and perhaps the two most famous characters in fiction decided to cast their lot together and locate on 221B Baker Street.

The staid Dr. Watson, after living with Holmes several weeks, took it upon himself to make an appraisal of his unusual roommate. His conclusions were that Holmes’ knowledge in certain areas, such as literature, philosophy, and astronomy, was “nil,” but in the field of chemistry, “profound.”

Holmes as a college student had evinced a marked interest in chemistry. When telling Dr. Watson the story of *The “Gloria Scott,”* his first case, he said: “All this occurred during the first month of the long vacation. I went up to my London rooms, where I spent seven weeks working out a few experiments in organic chemistry.”

Holmes maintained his interest in chemistry throughout the years. In *The Red-Haired League*, Watson describes how he called on Holmes one day: “A formidable array of bottles and test-tubes, with the pungent cleanly smell of hydrochloric acid, told me that he had spent his day in the chemical work which was so dear to him.” When Watson asked him whether he had reached a solution, Holmes answered in the affirmative, and stated it was the bisulphate of baryta. Watson, evidently disgusted said, “No, no, the mystery.” Holmes’ reply was: “Oh, that! I thought of the salt that I have been working upon.” Today one seldom hears the word “baryta” in this country; we speak of it as “barium.” The bisulphate of baryta, therefore, which Holmes mentioned, becomes barium sulphate.

The hobby of the great detective, it appears, afforded him relaxation, for we find:

“Well, I gave my mind a thorough treat by plunging into a chemical analysis. One of our greatest statesman has said that a change of work is the best rest. So it is. When I had succeeded in dissolving the hydrocarbon which I was at work at, I came back to our problem of the Sholtos, and thought the whole matter out again.”

The Sign of the Four

In his later years, Holmes must have missed his chemical experiments; for once, when he was away from home, we find Watson writing that his friend’s temper was on edge because he especially yearned for his chemistry bench and his scrapbook (*The Adventure of the Three Students*).

Holmes was wont at times to announce the results of his chemical researches in a dramatic manner. According to Watson, his hero was not averse to seeking the limelight. We already have seen how Holmes acted when he discovered an agent precipitated only by hemoglobin. He insisted upon telling of his important discovery before giving Stamford an opportunity of presenting Watson. We find another typical example in *The Naval Treaty*:

Holmes was seated at his side-table clad in his dressing-gown, and working hard over a chemical investigation. A large curved retort was boiling furiously in the bluish flame of a

Bunsen burner, and the distilled drops were condensing into a two-litre measure. My friend hardly glanced up as I entered, and I, seeing that his investigation must be of importance, seated myself in an armchair and waited. He dipped into this bottle ... and finally brought a test-tube containing a solution over to the table. In his right hand he held a slip of litmus paper.

“You come at a crisis, Watson,” said he. “If this paper remains blue, all is well. If it turns red, it means a man’s life.” He dipped it into the test-tube and it flushed at once into a dull, dirty crimson. “Hum, I thought as much!”

It is hardly necessary to point out that such dramatic chemical discoveries so glibly announced by Holmes are rare, and it is not given to any one individual to find many of them in a lifetime.

Reference already has been made to the fact that the reader rather expects Dr. Watson to make mention of the chemical bench when he describes the rooms at 221B Baker Street. One imagines Dr. Watson of a winter evening sitting in his rocking chair before the fire, absorbed in a treatise on surgery while Sherlock Holmes is busily engaged with some chemical problem. In *The Adventure of the Mazarin Stone*, we find Watson writing that it was pleasant to find himself again at 221B Baker Street, and that he felt at home when he saw the old chemical bench, the chemicals, and the other paraphernalia in the living room. Watson again mentions this in *The Adventure of the Empty House*, where he speaks of the old landmarks, and especially of the chemical corner.

On one occasion, Watson described a small chemical laboratory in a dwelling place.

It appeared to have been fitted up as a chemical laboratory. A double line of glass-stoppered bottles was drawn up upon the wall opposite the door and the table was littered over with Bunsen burners, test-tubes and retorts. In the corner stood carboys of acid in wicker baskets. One of these appeared to leak, or to have been broken, for a stream of dark-coloured liquid had trickled out from it, and the air was heavy with a peculiarly pungent, tarlike odor.

The Sign of the Four

This passage shows clearly that Dr. Watson was quite familiar with the arrangement in chemical laboratories.

Previously, it was mentioned that at their first meeting Holmes had warned Watson that he preferred to do chemical experiments in his rooms. The good doctor probably did not think much about it at that time, but later he had occasion to get firsthand information. In *The Musgrave Ritual*, Watson, telling how untidy Holmes was, states, “Our chambers were always full of chemicals....” We find further, in *The Sign of the Four*:

He [Holmes] would hardly reply to my questions and busied himself all the evening in an abstruse chemical analysis, which involved much heating of retorts and distilling of vapors, ending at last in a smell which fairly drove me out of the apartment. Up to the small hours of the morning I could hear the clinking of his test-tubes which told me that he was still engaged in his malodorous experiments.

And again, in *The Adventure of the Dancing Men*, we find Watson writing that Holmes was fouling the air with some malodorous chemical experiment. And finally, in *The Adventure of the Dying Detective*, Watson relates that the long-suffering Mrs. Hudson, too, often had to put up with noxious odors emanating from Holmes’ chemical experiments.

One wonders whether on occasion Watson regretted the day he agreed to share an apartment with his famous friend. It may be that Watson, who also had had considerable chemistry in his medical student days, did not mind too much the ill smells which from time to time polluted their living quarters; at any rate, there is no indication that Watson ever thought of seeking other quarters on this account.

Not only did Holmes’ chemical experiments fill the apartment with unpleasant odors, but he also kept ungodly hours. In *The Adventure of the Copper Beeches*, we find:

The telegram ... came late one night, just as I was thinking of turning in, and Holmes was settling down to one of those all-night chemical researches which he frequently indulged in, when I would leave him stooping over a retort and a test-tube at

night and find him in the same position when I came down to breakfast in the morning.

This instance, as well as others just mentioned, clearly show that Holmes was apt to devote long hours to his experiments. Evidently, he thought nothing of sitting up until the small hours of the morning, or, for that matter, all night, pursuing his researches. If young college instructors in departments of chemistry showed such zeal, they surely would soon gain recognition in their fields. Working all day and most of the night, however, probably is too much to expect of the average human frame.

The casual reader naturally takes it for granted that Holmes' love for chemistry was due to his interest in crime detection; that is, he intended to use the knowledge he gained from his experiments for practical purposes. We recognize this today as applied research. I would like to emphasize that Holmes was interested in pure or basic research, too—in other words, research which has no immediate practical value. Instances may be found in the tales to support this view. I will mention only two: “That will do very nicely. Then perhaps I had better postpone my analysis of the acetones, as we may need to be at our best in the morning” (*The Adventure of the Copper Beeches*). And, in *The Final Problem*, Holmes remarks rather plaintively to Watson: “Between ourselves, the recent cases in which I have been of assistance to the royal family of Scandinavia, and to the French Republic have left me in such a position that I could continue to live in the quiet fashion which is most congenial to me, and to concentrate my attention upon my chemical researches.”

His interest in basic research is rather remarkable, for in that day there were comparatively few persons engaged in any type of chemical research. Today, of course, there are thousands of chemists working in institutions of higher learning and also in many industries. Indeed, some of the chemists in large industrial plants are not only allowed but encouraged to do basic research.

We can conceive of Holmes as a famous professor of chemistry in a great university. In *The Adventure of the Dancing Men*, Watson writes that Holmes had propped his test tubes in a rack, and that he then began to lecture like a professor addressing his class. As a teacher, he might at times have been somewhat curt with his students and probably impatient with the

stupid ones, but at all times he would have been fair. He had the energy, stamina, and enthusiasm not only to be a stimulating teacher but a productive scholar as well. He would not have been an armchair professor, but rather a leader in his chosen field. We are thankful that he did not choose an academic career, for then the delightful Sherlock Holmes stories never would have been written, and the world would be a less interesting place in which to live.

In closing, we can paint a pleasant picture of Sherlock Holmes in his retirement in his comfortable home on the Sussex Downs. We can imagine him in the long English twilight working at his chemical bench before a large window overlooking the beautiful countryside. His fingers are stained with acids, alkalies, silver nitrate, and other potent chemicals. His pipe is going full blast, and his brow is probably perplexed as he scans a recent learned treatise on the chemistry of the aldehydes or the alkaloids.

While his researches may be profound and manifold, it is more than likely that he devotes his best efforts toward the detection of some obscure poisons, with the hope that some scheming sinister criminal may be brought to justice. Time does not hang heavily on his hands in his retirement, for there is no end to chemical researches; infinite problems present themselves to the prepared mind. With his keen intellect, his wide knowledge, and his rich imagination, he has doubtless outlined investigative work which will keep him happy and busy for years to come.

DOCTOR WATSON'S UNIVERSAL SPECIFIC

“... with the aid of ammonia and brandy, I had the satisfaction of seeing him open his eyes.”

The Greek Interpreter

The therapeutic agent most frequently used by Dr. Watson was brandy. Mention is made of this stimulant in a number of the tales. Let us examine the conditions in which Dr. Watson's favorite remedy was used.

In *The Adventure of the Blue Carbuncle*, Sherlock Holmes had accused the pathetic Ryder of stealing the valuable carbuncle. The poor wretch turned pale, and Holmes remarked to Watson, “Give him a drink of brandy.” In describing this incident, Watson writes, “For a moment he had staggered and nearly fallen, but the brandy brought a tinge of colour into his cheeks....”

We find, in *The Adventure of Wisteria Lodge*, that when the eminently respectable Mr. Scott Eccles was relating his strange experience to Sherlock Holmes and Dr. Watson, Inspector Gregson of Scotland Yard came in unannounced and informed Eccles that Mr. Garcia, his host of the preceding night, had been found murdered. Dr. Watson writes that their client turned deathly pale. Holmes quickly suggested to Watson that he give Eccles a brandy and soda. This evidently helped the poor fellow; he gulped it down, and his face soon resumed its normal color.

Dr. Watson, in the story of *The Greek Interpreter*, took credit for saving the life of Mr. Melas, who was found locked in a room filled with a poisonous gas. Holmes and Watson rushed into the room and dragged out the victim. Dr. Watson writes that, “... with the aid of ammonia and brandy, I had the satisfaction of seeing him open his eyes.”

In *The Naval Treaty*, Holmes found the highly important document which his client Mr. Percy Phelps thought had been lost through his own negligence. The great detective invited him to have breakfast with him and Watson; and when Phelps took the lid off the dish which was served him,

and saw the precious papers lying there, he became hysterical with joy: “Then he fell back into an armchair so limp and exhausted with his own emotions that we had to pour brandy down his throat to keep him from fainting.”

We learn, in *The Adventure of the Priory School*, that when Thorney Huxtable, M.A., Ph.D., made his dramatic entrance and collapsed upon Holmes’ bearskin hearthrug, both the detective and the medical man extended helping hands. Holmes hurriedly placed a cushion under his head, and Watson got ready to give him a drink of brandy. However, the pompous fellow fell to the floor in a faint. When he regained consciousness, he asked for some milk and a biscuit. This revived him, and he then told his story. We do not know whether he was an abstainer or not.

Brandy evidently enjoyed a reputation as an effective stimulant even among laymen. In perusing the tales, we find in one instance that it was given by a tutor to his servant who had suffered a mental shock; and in another, a constable intended to give it to a beautiful young woman who, he thought, had fainted. Dr. Watson relates, in *The Adventure of the Three Students*, how Mr. Soames, tutor and lecturer at one of the great British universities, administered brandy to his servant, Bannister. The highly nervous tutor thought his man had disturbed the examination papers on his desk; and when Bannister saw that the papers had been disturbed, he appeared as if he were about to faint. Soames immediately gave him a drink of brandy; but, in spite of the stimulant, Bannister appeared to be in a state of collapse, and threw himself into a nearby chair. We shall never know whether or not it was an actual collapse, the brandy failing to help, for later it came out that the servant was desirous of concealing something on the seat of the chair. We may safely assume, however, that in any event Bannister welcomed the proffered drink.

We learn from the pen of Dr. Watson, also, how Lady Hilda duped Constable McPherson and obtained the precious blue envelope which contained high state secrets (*The Adventure of the Second Stain*). The crestfallen constable told Holmes that, when Lady Hilda had seen the stain on the carpet, she apparently fell down in a faint. The constable rushed outside to a nearby corner to procure some brandy, but when he returned, the noble lady had disappeared. We wonder whether the constable treated himself to the drink which he had procured for her.

In two other instances—one related by Sherlock Holmes and the other by Dr. Watson—brandy was administered with a lavish hand indeed, and the inference may be drawn that the subjects probably drank about all they could decently manage. In *The Adventure of the Lion's Mane*, Ian Murdock, who had come into contact with the jellyfish *Cyanea capillata*, staggered into the room, on the verge of a collapse, crying for brandy. Holmes, in writing this tale, tells how he administered a half-tumbler of the stuff. The treatment appeared to be of some help, but the patient was still in great pain. Holmes gave him several more large drinks. Finally, Murdock's head fell upon the cushion in a state of unconsciousness. It was well that he did not attempt to get up and walk. I doubt whether he could have stood up.

In the second example, Victor Hatherly, a hydraulic engineer, had suffered a terrifying experience (*The Adventure of the Engineer's Thumb*). In his effort to escape being crushed to death by the powerful hydraulic press, he had torn off his thumb. When he presented himself in Dr. Watson's surgery and began to tell of his ghastly experience, he became hysterical. Dr. Watson, who had given him a drink of water previously, rose to the occasion: "I dashed some brandy into the water, and the colour began to come back to his bloodless cheeks." He then proceeded to bandage the area where the thumb had been. This must have been a most painful operation indeed, and it is hoped that Dr. Watson did not spare the brandy. There is reason to believe that he did not. Since the engineer wished to report the incident, Watson took him to consult with Sherlock Holmes. The latter listened to the engineer's story, "... placed a pillow beneath his head, and laid a glass of brandy and water within his reach." Dr. Watson did not indicate whether the patient was able to go home under his own power.

Let us turn now to the longer tales of Sherlock Holmes. In *The Sign of the Four*, when Jonathan Small was telling his story to Holmes and his colleagues, the astute detective remarked to him: "... you had best take a pull out of my flask, for you are very wet." And a little later: "He stopped and held out his manacled hands for the whiskey and water which Holmes had brewed for him."

In *The Hound of the Baskervilles*, the reader will recall that the Baronet, Sir Henry, was attacked by the spectral hound and miraculously escaped death. Dr. Watson writes: "Lestrade thrust his brandy flask between the Baronet's teeth, and two frightened eyes were looking up at us." And, when

Holmes asked him whether he was strong enough to stand, the Baronet replied: "Give me another mouthful of that brandy and I shall be ready for anything." Brandy was given also to the beautiful Mrs. Stapleton, the villain's wife, when she was found in her house in a pitifully exhausted state.

Dr. Watson himself had need of brandy as a stimulant at least once in his lifetime. He, like everyone else, believed that Sherlock Holmes had died with Professor Moriarty at the Reichenbach Falls. We cannot blame the doctor for fainting when, three years later, he saw Holmes in the flesh standing on the opposite side of the study table. Watson claims that this was the only time in his life he had ever fainted. When he regained consciousness, he found that Holmes had undone his collar and had administered brandy to him (*The Adventure of the Empty House*).

Brandy and whiskey are, of course, used medicinally today, but probably not as frequently as several decades ago. Some scientists argue that alcohol does not cause a physiologic stimulation but rather a depression. Pharmacologically, this is true, for it is now accepted that the talkativeness and hilarity so often produced by alcohol do not indicate a true stimulation of the nervous system but a removal of the inhibition which the higher centers normally exert.

On the other hand, it is still held by many physicians, and some pharmacologists as well, that brandy and whiskey are useful therapeutic agents in emergencies. Dr. Torald Sollman, dean of American pharmacologists, feels that alcohol is a quick-acting stimulant, and has a definite place in treating conditions such as syncope, exhaustion and certain forms of shock. He attributes its effect to reflex stimulation, which accounts for its brief action.^[4] He suggests that a dose of 25 cc. (somewhat less than an ounce) of whiskey or brandy be given, and that it be repeated every ten or fifteen minutes.

Dr. Watson, throughout his stirring adventures with Sherlock Holmes, was called upon to meet many emergencies. He was presumably justified in resorting to frequent administrations of brandy. He had ample opportunity for observing the effects of his medication, and apparently was well satisfied with his results, for he continued to employ brandy as a stimulant throughout the years. One practical justification for doing so was that it

seemingly always was available. There is every reason to believe that Dr. Watson carried a flask of brandy, as did Sherlock Holmes. Even Lestrade, the dependable Scotland Yard detective, was not found wanting when the occasion demanded, as we have seen in the story of *The Hound of the Baskervilles*. Apparently, it was the practice of the day.

In conclusion, we may rest assured that Dr. Watson was firmly convinced that his favorite remedy, brandy, as he employed it, helped alleviate the suffering of his unfortunate fellow men. We may accept, also, that he administered it only in an emergency, and when he was morally certain that there was a real indication for its use—in short, he observed faithfully the ethics of his noble profession.

DOCTOR WATSON, ENDOCRINOLOGIST

“The emotional qualities are antagonistic to clear reasoning.”

The Sign of the Four

The colorful story of *The Adventure of the Creeping Man* has always especially interested me, for the theme embraces the fascinating field of endocrinology. The plot revolves around Professor Presbury, the Camford physiologist, who is portrayed as a biological scientist of international renown. It is in order to review briefly the setting of the story.

Professor Presbury, a widower in his early sixties, had fallen madly in love with a very young girl, the daughter of one of his colleagues. During the course of the love affair, he began to show pronounced changes in personality; he became secretive, highly irritable at intervals, and on occasion even savage. When this portly and dignified physiologist began to do such astounding things as walking on all fours and swinging on the ivy-covered walls with the abandon and agility of an ape, he naturally caused grave concern to his only daughter Edith and to his associate Mr. Bennett, Edith's fiancé. Furthermore, the professor's heretofore faithful wolfhound Roy had tried to attack his master on several occasions. Events came to such a pass that Mr. Bennett and Miss Presbury asked Sherlock Holmes to make an investigation.

Holmes summoned Dr. Watson to 221B Baker Street one night in September, and casually announced that he was seriously considering writing a monograph on the uses of dogs in detective work. He observed that the action of a dog often reflects family life, and gave as an illustration the case on which he was presently working. He believed that the changed behavior of the wolfhound toward his master might prove to be an important clue in solving the mystery. In point of fact, this eventually proved to be true.

During this discussion, Mr. Bennett called on Sherlock Holmes and, in the presence of Dr. Watson, reviewed the strange actions of the professor and the dog's attitude toward his master. He brought out the important

information that it was every ninth day when Presbury acted so abnormally. With this latter fact in mind, it was not difficult for Sherlock Holmes to arrange to be on hand at a time when the professor would probably perform again his astonishing antics.

Accordingly, one night Holmes and Watson secreted themselves in the shrubbery in the professor's extensive yard and awaited events. Holmes had previously instructed Mr. Bennett to follow secretly his future father-in-law should he venture out on this particular night. Their vigil was rewarded, for about midnight the famous physiologist emerged from his house clad in his dressing gown. To the astonishment of his onlookers, he began walking on all fours. Dr. Watson describes his further antics. The professor, with surprising agility, began to ascend the ivy-colored walls, springing aimlessly from branch to branch, but apparently hugely enjoying his singular powers. He finally tired of thus disporting himself, came down to the ground, and moving along on his hands and feet, made his way to the barn where Roy was tethered. Remaining at a safe distance, he deliberately teased the dog until the infuriated beast, having managed to slip his collar, sank his teeth into the neck of the unfortunate professor. The two onlookers, with the help of Bennett, rushed to the rescue; without their intervention, the dog doubtless would have killed his master.

Holmes had been working quietly on the case and had discovered that Presbury, in his desire to regain the vigor of his youth, had connived with an obscure Prague scientist who was attempting to unravel the secret of rejuvenescence. Holmes found that this individual was attacking the problem by studying the physiologic effects on man of the administration of serum obtained from certain apes. The serum he had prescribed for Professor Presbury had been prepared from the langur, a large black-faced monkey which lives on the slopes of the Himalayas. Holmes emphasized that the langur was both a climber and a crawler. The implication is obvious; it accounts for the professor's mode of progression and his agility as a climber.

Holmes, discussing the case later with Watson, remarked that it was the untimely love affair with the young girl which had given the professor the idea of trying to turn himself into a younger man. Holmes also commented on the strange behavior of the wolfhound; he felt the dog had turned on his master because he thought it was a monkey he was attacking.

The theme of this story is of interest particularly to biologically trained men. About the time it was written, there was a great deal of interest in the possibility of rejuvenation by the transplantation of testicular glands or by an injection of their extract. The work of Steinach and of Voronoff, the European physiologists, had received immoderate publicity.

The real pioneer in the study of rejuvenation was, of course, Charles-Edouard Brown-Séquard (1818-94), who reported the results of his experiments in June 1889 before the *Société de Biologie*. This brilliant French physiologist during the course of the preceding fortnight had administered to himself six subcutaneous injections of the water extract of ground-up testicles of dogs or of guinea pigs. He was then seventy-two years old. He enthusiastically reported that the effects had been astonishingly beneficial, and firmly believed that both his physical forces and his mental activities were enormously improved. He claimed that he felt less fatigued at the end of the day, which was quite important to him, since he spent long hours in his laboratory.

Brown-Séquard had such an enviable reputation as a scientist that testicular extracts at once became widely and indiscriminately used. The idea spread like wildfire. More than twelve thousand physicians were administering testicular extracts to their patients before the end of the year. Some of the more conservative physicians in England and in the United States, however, were highly critical of Brown-Séquard's recent researches. They spoke derisively of the extract as "senile folly," "elixir of youth," and made other uncomplimentary remarks. In fairness to Brown-Séquard, it should be stated that he realized the extract was not a panacea for all illnesses. But he did feel, up to the time of his death, which occurred four years later, that testicular extracts built up what he termed "nervous force."

It is now recognized by modern endocrinologists that the distinguished Brown-Séquard had not controlled his experiments well. The testicular extracts he used were probably inactive. If I may be technical about it, he had made an aqueous extract which would leave behind the hormones, which are fat soluble. It is known also that some of the claims made by Steinach and by Voronoff were grossly exaggerated.

It is possible that people who are not familiar with the science of endocrinology might gain the impression that the administration of ape

serum could produce the effects so vividly described in *The Adventure of the Creeping Man*. The concept that such a serum exists is, of course, rank nonsense. Even though the tale is incredible, I confess that I have always enjoyed reading it. This may be because it is a story about a physiologist—and an unusual physiologist, for he was also extremely wealthy. Parenthetically, mention should be made that the story actually touches on science-fiction, which is at present so much in vogue.

The theme of rejuvenation has been used by numerous authors. One story which comes to mind is the novel *Black Oxen*, written by the late Gertrude Atherton, and published many years ago. This book was widely read. The story, of course, was actually in the nature of science-fiction.

The subject of the renewal of youth is not as popular as it was some years ago. There are still a few charlatans in the medical profession who take advantage of the public and prescribe gonadal therapy irrationally and even indiscriminately. These misguided medical men make wild promises as to the permanent benefits to be derived from administration of gonadal extracts. Indeed, there have been a few such doctors in the United States, and one, at least, is reported to have amassed a fortune. A few of these men have acted in good faith, and we may charitably assume they believed that the administration of gonadal extracts stimulated the libido of their patients or enabled them to lead a more strenuous life. Be that as it may, many such practitioners have acted on slender evidence, and they surely have, in a large measure, the will to believe.

Since the time Dr. Watson wrote *The Adventure of the Creeping Man*, there have been highly important developments in gonadal therapy. Testosterone, for example, a preparation from the male gonads, is presently widely used and has a definite place in the therapeutic armamentarium of any reputable physician. There are other preparations, such as the estrogenic substances and progesterone, which have therapeutic uses. These latter substances are extracted from the female gonads. It would take us too far afield to elaborate on these or to give the indication for their uses. There is good reason to believe that there will be many new developments in this particular field, for the science of endocrinology is still in its infancy.

GENETICS AND SHERLOCK HOLMES

“The point under discussion was how far any singular gift in an individual was due to his ancestry and how far to his early training.”

The Greek Interpreter

Allusions are made in several of the tales to the subject of genetics—that is, the science of heredity. Many of the observations set forth, although perhaps not entirely acceptable today, are nevertheless intellectually stimulating.

Nearly all human beings—some much more than others—are interested in matters pertaining to heredity. Most of us like to hear about our grandparents and even about our great-grandparents, despite the fact that the deeds of some of our ancestors, and the lives they lived, may cause us some embarrassment.

In this connection, I am reminded of a statement made to me one evening (a long time ago) by a distinguished Southerner of the old school, a Kentucky gentleman, during a discussion of our ancestors. In effect, he suggested that perhaps we should not examine the lives of our early kinfolk too closely, because one of them might have been hanged as a horse thief.

When we uncover some irregularity in the behavior of our ancestors, we often toss it aside by pointing out that after all they were products of their times, which indicates, in part at least, that environment was highly important. There is doubtless considerable truth in this statement, but an honorable man in the eighteenth century probably had (or anyway, *should* have had) the same high standard of conduct as an honorable man in the twentieth century. Rather than sermonize further, let us turn to the tales of Sherlock Holmes.

We find Doctor Watson and Sherlock Holmes talking together on a beautiful summer evening. Doctor Watson writes:

... the conversation ... came round to the question of atavism and hereditary aptitudes. The point under discussion was how far any singular gift in an individual was due to his ancestry and how far to his early training.

The Greek Interpreter

Watson expressed the thought that Holmes' faculty of observation and his peculiar facility for deduction were due to his own systematic training. Holmes ventured this was true only to some extent. When Watson asked him why he thought it hereditary, Holmes replied, "Because my brother Mycroft possesses it in a larger degree than I do."

Doctor Watson was amazed at this answer, and expressed the sentiment that if any other man in England possessed greater keenness in observation and deductive powers than did Sherlock Holmes, the public would surely know of him. Watson hinted it was Holmes' modesty which caused him to extoll his brother's powers. Holmes quickly made this rejoinder: "I cannot agree with those who rank modesty among the virtues. To the logician all things should be seen exactly as they are, and to underestimate one's self is as much a departure from truth as to exaggerate one's own powers. When I say, therefore, that Mycroft has better powers of observation than I, you may take it that I am speaking the exact and literal truth."

There are doubtless many people who would take issue with the thoughts Holmes expressed. But they do contain more than a modicum of truth. Is there not an expression to the effect that false modesty is not a virtue? Does not the matter of intellectual honesty also enter into this picture? We will not at this time pursue the matter further, although it is tempting.

Sherlock Holmes suggested to Dr. Watson that, since he had never met Mycroft, it would be fitting for him to make his acquaintance. Accordingly, they set out for the Diogenes Club, where Holmes assured Watson that Mycroft, on account of his routine life, would be found this very hour of the day.

Watson was presented to this extraordinary person:

... his face ... had preserved something of the sharpness of expression which was so remarkable in that of his brother. His

eyes, which were of a peculiarly light, watery gray, seemed to always retain that far-away introspective look which I had only observed in Sherlock's when he was exerting his full powers.

Watson has skillfully depicted the family likeness—that is, the highly intelligent face and the dreamy and introspective look. Obviously, Mycroft and Sherlock were both remarkably keen, alert men.

At the Diogenes Club, Mycroft and Sherlock seated themselves before a window which gave them a clear view of the street. As they sat there conversing, two strangers were approaching along the sidewalk. Mycroft called attention to them, and the following brilliant dialogue ensued:

“Look at these two men ... coming toward us....”

“The billiard-marker and the other?”

“Precisely. What do you make of the other?”

(Dr. Watson then explained that he noted some chalk marks on the waistcoat of one of the men which was the only evidence of billiards he could see, and that the other one was carrying several packages.)

“An old soldier, I perceive,” said Sherlock.

“And very recently discharged,” remarked the brother.

“Served in India, I see.”

“And a non-commissioned officer.”

“Royal Artillery, I fancy,” said Sherlock.

“And a widower.”

“But with a child.”

“Children, my dear boy, children.”

Watson laughingly remarked that this was too much. Sherlock and Mycroft assured him it was not. They pointed out some of the salient points which had enabled them to make their astute conclusions. Among those mentioned were that the man in question had the bearing of a soldier, wore the expression of authority, and had a deeply tanned skin (which suggested

he had served in India). The fact that he was wearing deep mourning indicated he was probably a widower.

The dialogue actually was a contest of wits between Sherlock and Mycroft. The latter obviously won when he said, “Children, my dear boy, children.” This winning stroke Mycroft explained in this way: “He has been buying things for his children.... There was a rattle which shows one of them is young ... a picture book under his arm shows that there is another child to be thought of.”

When Mycroft had finished, Holmes merely turned to Watson and smiled, not only to acknowledge his brother’s genius but also his superiority in observation and deduction.

Sherlock Holmes apparently believed firmly that heredity played an extremely important part in the life of an individual. There is no end to the argument among biologists whether heredity or environment plays the more significant role in the development of an individual. It is especially so with a person’s mental capacity. One who unfortunately has inherited a mediocre mind will always be handicapped, to a degree, regardless of his environment or training. The mental powers will remain ordinary. Conversely, the individual who is endowed with a keener intellect will always be potentially the intellectual superior of his less fortunate brother. There is no doubt that training and environment in general will aid both types of individuals, but even the most ideal developmental conditions can not completely overcome a deficient genetic constitution.

In the story of *The Yellow Face*, it will be recalled that a white woman had married a Negro. The child, according to the story, had even a darker skin than her father: “It was our misfortune that our only child took after his people rather than mine. It is often so in such matches, and little Lucy is darker than her father was.” This statement must be challenged. It appears incredible that a child born of a white mother could be darker than her Negro father. If the mother had had some Negro blood, then by certain arrangements of genes this could have happened, but not otherwise. Dr. Watson doubtless is in error in this instance.

A hereditary likeness once enabled Holmes to solve a mystery (*The Adventure of the Cardboard Box*). A maiden lady (Miss Cushing) had received in the mail a small cardboard box containing two human ears—one

that of a woman. Holmes was called in, and during the course of the investigation visited the receiver of this gruesome package. He noted the resemblance of one of the severed ears to the ears of the lady upon whom he was calling. Because of the striking likeness, he felt certain that the person whose ear had been dismembered was a close relative of the Miss Cushing whom he had come to interrogate. Holmes remarks to Dr. Watson: "I perceived that her ear corresponded exactly with the female ear I had just inspected.... There was the same shortening of the pinna, the same broad curve of the upper lobe, the same convolution of the inner cartilage.... It was evident that the victim was a blood relation, and probably a very close one." It will be recalled that the dismembered ear proved to be that of Miss Cushing's sister.

Holmes propounds an interesting theory of the development of the individual, in *The Adventure of the Empty House*. Speaking of the terrible Colonel Moran, he expressed the idea that the individual represents the whole procession of his ancestors in his development, and that the person becomes the epitome of the history of his own family. Doctor Watson thought this rather fanciful, and Holmes' rejoinder was that he would not insist upon it.

In *The Hound of the Baskervilles*, Holmes successfully demonstrates by means of a family portrait that the villain Stapleton is actually a Baskerville. It will be remembered that Holmes stands upon a chair and uses his arm to cover the broad hat and the long ringlets depicted in the portrait. Now that only the features are visible, the portrait of old Hugo Baskerville boldly stands out. Watson is astounded when he beholds the results. Holmes remarks that it was an interesting instance of a throwback, apparently both physical and spiritual.

The discovery that Stapleton markedly resembles one of his early ancestors is not too surprising, for this surely could happen in any family. The remarkable thing is that Holmes had had the acumen to cover up certain portions of the old family portrait to make the face of old Hugo Baskerville stand out. This clever demonstration would not have occurred to many people. The extraordinary family resemblance portrayed is a splendid example of atavism—that is, a reappearance of the characteristics of a remote ancestor which presumably had been more or less absent in intervening generations. The matter of atavism would make a most

interesting study. Therein lies the value and interest of the old family album which used to grace the center table in the parlor.

In *The Adventure of the Copper Beeches*, Holmes makes a neat point in regard to the subject of heredity: “My dear Watson, you as a medical man are continually gaining light as to the tendencies of a child by the study of the parents. Don’t you see that the converse is equally valid. I have frequently gained my first real insight into the character of parents by studying their children.”

This is an interesting analysis. Let us pursue the thought further. The expression is often heard that a child’s behavior resembles that of one of his parents. A popular way of putting it is that he is “a chip off the old block.” This makes it appear that the pattern of behavior is set up early in life. An old maxim has it: “As the twig is bent, so the tree will incline.”

The problem is not always as simple as Holmes suggests. It is true that we all have seen fine and lovable children whose parents have similar qualities. But we also have seen ill-mannered, nasty-tempered children whose parents have fine characters and possess charming manners. One is reminded of a statement attributed to the gentle and amiable Robert Louis Stevenson: “It has been a source of perpetual mystification to me where all the disagreeable medical students go to, and all the admirable doctors come from.”

I think the same may be said about some of the spoiled, unlovable children. Many of them develop into sterling young men and women who later in life carry on with noble purpose and high resolve. These individuals often form the backbone of their community.

The entire matter of heredity is a complicated one, and much is still to be learned about the subject. It is extremely important in the study and practice of medicine. There are some common diseases which certainly run in families, such as migraine, high blood pressure (hypertension), and diseases of the coronary arteries. And it is definitely known that diabetes may be inherited. In my own experience, I have known a father with diabetes whose three daughters had all developed this condition early in life. There are other diseases just as formidable: for example, certain neurological and mental diseases. Any researches which may throw any light on the subject of heredity should be encouraged.

As we have seen, there are not many allusions in the Sherlock Holmes stories to genetics. It is a pity that the topic of crime and heredity was not touched upon. As far as I know, no specific reference is made to it. I am not sufficiently acquainted with crime statistics to know how commonly one would find several criminals in one family—that is, in a single generation. However, one does think of those Missouri bandits: the James brothers, the Younger brothers, and the Daltons. Perhaps there are others. In any case, one would like to know whether their parents or grandparents had criminal records. I submit, however, that these last-named cases are rather special ones, for the James brothers and their colleagues were products of an unusual period. Doubtless, a black sheep may be found in an otherwise respectable family. As Sherlock Holmes stated to Dr. Watson, once in a while a carrion crow appears among the eagles (*The Adventure of Shoscombe Old Place*).

THE ZOOLOGICAL DOCTOR WATSON

“The coyote skulks among the scrub, the buzzard flaps heavily through the air, and the grizzly bear lumbers through the dark ravines....”

A Study in Scarlet

Dr. Watson was a man of catholic tastes and diversified interests. Not only was he a capable and experienced physician, but he was a man of letters, an amateur detective, and an athlete of considerable ability. No one, as far as I know, has emphasized his interest in zoology. This trait is more apparent to those acquainted with the biological sciences, but even the most casual reader cannot but notice the frequent references to animal life.

The plots in a number of the tales depend entirely upon a zoologic background. In *The Adventure of the Lion's Mane*, for example, the death of a person as well as that of a dog are attributed to the poisonous sting of *Cyanea capillata*, a huge jellyfish. A swamp adder, which Watson describes as “the deadliest snake in India,” is used to kill two people in *The Adventure of the Speckled Band*. In another tale, the plot hinges upon the fact that a goose has swallowed a rare jewel, which is eventually recovered from its crop (*The Adventure of the Blue Carbuncle*). A race horse is answerable for the mysterious death of his trainer (*Silver Blaze*). In *The Adventure of the Veiled Lodger*, a lion is responsible for the dilemma of the unfortunate lodger. The central theme in *The Adventure of the Creeping Man* has to do with the antics of a scientist who at times shows apelike qualities. Finally, in *The Hound of the Baskervilles*, the grisly spectral hound plays the title role.

Let us consider first some references to the lower forms of life, such as insects. The naturalist Stapleton, in *The Hound of the Baskervilles*, was supposed to be an authority on butterflies and moths, that is, lepidoptera. When he and Watson met one day on the moor, the naturalist invited him to see his collection, which he described as the most complete in the southwest of England. Later in the story, when searching the house for the fiendish

Stapleton, Dr. Watson describes a room fitted up as a small museum with glass-topped cases filled with butterflies and moths.

The scholarly recluse Nathan Garrideb evidently was another such collector, for when Holmes and Watson visited his house they found cases of butterflies and moths on each side of the entrance (*The Adventure of the Three Garridebs*). One must remind the reader that butterflies are distinguished from moths by the shape of their antennae, the former having clubbed antennae, the latter not.

Two allusions are made to the antennae of insects. In *The Adventure of the Illustrious Client*, the waxed tips of the Baron's mustache are likened to an insect's antennae. The other is found in *The Hound of the Baskervilles*. Dr. Watson, observing Dr. Mortimer, the country practitioner, rolling a cigarette, remarks that his long and quivering fingers remind one of the agility and restlessness of the antennae of an insect. These descriptions are extremely apt, since the antennae of insects are nearly always in motion.

In *The Adventure of Black Peter*, Captain Peter Carey was found one morning harpooned to death in a small shed near his house. Holmes remarks that the captain had been pinned to the wall like a beetle on a card.

Let us go higher in the zoological scale. It is evident that Holmes despised snakes, for he refers to them as extremely repugnant creatures and remarks to Watson that watching them in a zoo gives him a most unpleasant sensation (*The Adventure of Charles Augustus Milverton*). A number of animals which live in the water are mentioned: namely, fish, jellyfish, sharks, seals, and crocodiles. For instance, when Holmes introduces his brother Mycroft to Watson, the latter writes that Mycroft put "... out a broad fat hand like the flipper of a seal" (*The Greek Interpreter*). And in *The Sign of the Four*, Jonathan Small had the extreme misfortune of having his leg bitten off by a crocodile.

Birds are not neglected. In *The Adventure of the Solitary Cyclist*, Watson tells how he and Holmes on one of their walks enjoyed the music of the birds. On another occasion, he likens Holmes to a lank bird (*The Adventure of the Dancing Men*). Watson, in *The Adventure of the Priory School*, speaks of the plover and the curlew; and in *The Hound of the Baskervilles*, the raven and bittern are used effectively to accentuate the utter loneliness of the moor. The ravens croak loudly behind the tor, and this is the only life

stirring over the vast landscape. When Watson one day asks the naturalist Stapleton what had caused the bloodcurdling sound on the moor, the latter's shifty reply is that they had heard the cry of the last of the bitterns. In *A Study in Scarlet* occurs the felicitous expression: "... the buzzard flaps heavily through the [desert] air..." In *The Adventure of Shoscombe Old Place*, Holmes makes the pertinent statement, when speaking of the weakness of Lord Robertson, that even among eagles one may find a carrion crow.

An occasional reference to the eggs of birds also occurs. Watson writes, "... a young lady entered the room with a bright quick face, freckled like a plover's egg ..." (*The Adventure of the Copper Beeches*). The cipher Holmes employed in *The "Gloria Scott"* contained the words, "pheasant hen eggs."

So much for the lower forms of life; let us move on to the mammals. Attention is called to several small mammals—in *The Adventure of the Crooked Man*, a mongoose; and in *The Sign of the Four*, a stoat and a mole as well as a weasel. The reader may recall that other names for a stoat are "ermine" or "ferret."

Large mammals are referred to more often. Dr. Watson writes, in *A Study in Scarlet*, "The coyote skulks among the scrub ... and the grizzly bear lumbers along the dark ravines..." Big-horn sheep also appear: "... there stood a creature somewhat resembling a sheep in appearance, but armed with a pair of gigantic horns. The big-horn—for so it was called..." In this same story, bison are also to be found. Holmes, exasperated when looking for footprints, says, "If a herd of buffaloes happened along, there could not be a greater mess." In still another instance, Watson writes, "... the observer would have come to the conclusion that one of those great herds of bisons which graze upon the prairie land was approaching him." A herd of bison is referred to again in *The Boscombe Valley Mystery*. Holmes feelingly remarks, "Oh, how simple it would all have been had I been there before they came like a herd of buffalo and wallowed over it!" It is obvious that Holmes was impressed with the immense herds of buffalo which once roamed over the great plains of our romantic West.

In *The Valley of Fear*, we come across an interesting analogy. Holmes remarks to Watson that Porlock, although unimportant in his own right,

nevertheless takes on importance through his close touch with the master criminal Professor Moriarty. He spoke of the relationship of the two men as that of the pilot fish to the shark, or of the jackal to the lion; that is, one was insignificant and the other formidable. In this same story, our attention is called to two other mammals—the wolf and the caribou. In *The Adventure of the Speckled Band*, we are introduced to two African animals, the cheetah (reputed to be the fastest animal on earth) and the baboon. And in *The Adventure of the Creeping Man*, reference is made to one of the higher apes, the langur, which is described as a large animal with a black face living on the slopes of the Himalayas.

When we read about large mammals, big-game hunting comes to mind, at least to the sportsman. Several allusions to this exciting and dangerous sport are found in the tales. In *The Naval Treaty*, we read, “... it has the sort of excitement about it that the sportsman feels when he lies beside the water course and waits for the big game.” In *The Adventure of Black Peter*, we read again of the hunter’s thrill while lying in wait for some unknown large beast to approach the water hole.

When Sherlock Holmes ironically introduces the terrible Colonel Moran to Dr. Watson and Lestrade, he says, among other things, that the Colonel is an outstanding big-game hunter, practically without peer, and that his bag of tigers remains unrivaled (*The Adventure of the Empty House*). In *The Adventure of the Bruce-Partington Plans*, Holmes notes that it was such a foggy day in London that a thief or murderer could roam the streets and pounce on his prey like a tiger in the jungle. In another instance, Holmes tells Watson about Count Sylvius, the famous big-game hunter who had stolen the Crown diamond. The nonchalant detective facetiously remarks that if the Count could add him (Holmes) to his bag of big game, it would be an outstanding triumph (*The Adventure of the Mazarin Stone*).

The numerous allusions to animal life in the tales show clearly that Dr. Watson was keenly interested in zoology. This is not unexpected, since he was a physician, and, as everyone knows, a student of medicine has a broad training in the biological sciences. It is doubtful whether a man untrained in biologic science could have used the references to animals as accurately and as effectively as did Dr. Watson.

Where did he obtain his love for zoology? Solely from his preparatory studies in medicine? Probably not. Dr. Watson was an athlete and an outdoor man. Such men frequently take a pronounced interest in plant and animal life, since it is before their eyes a good part of the time.

We may conclude, then, that Dr. Watson was not only a physician, an amateur detective, an athlete, and a man of letters, but a zoologist as well. In this age of specialization, it is refreshing to reflect that Dr. Watson had such a wide range of interests.

DOCTOR WATSON, CARDIOLOGIST

“Marsten had sprung out of his chair in a paroxysm of anger, when he suddenly pressed his hand to his side, his face turned a dusky hue, and he fell backward....”

The Sign of the Four

Dr. Watson did not profess to be a cardiologist, but rather a general practitioner of medicine. There are more allusions to minor surgery and to nervous disorders in the tales than there are to diseases of the heart or circulation. There are, however, several references to the latter which are of historical interest to medical students and practicing physicians.

Let us examine some of the references to cardiovascular conditions. In *The Sign of the Four*, a sudden cardiac death already cited above is described:

Marsten had sprung out of his chair in a paroxysm of anger, when he suddenly pressed his hand to his side, his face turned a dusky hue, and he fell backward, cutting his head against the corner of the treasure-chest. When I stooped over him, I found to my horror that he was dead.

The cause of death in this instance, so dramatically described, presumably was produced by a coronary occlusion. It is noteworthy that the patient was “in a paroxysm of anger” at the time he was stricken. One is reminded of John Hunter, the famous English anatomist, who in later life suffered from severe angina pectoris. He is reputed to have said that he was at the mercy of any rascal who saw fit to make him angry. Any intense emotion, of course, is capable of producing profound effects on the cardiovascular system.

It will be remembered in the novel, *The Hound of the Baskervilles*, that Sir Charles Baskerville had a cardiac disorder. His friend and physician, Dr. James Mortimer, had concurred that “Sir Charles’ health has for some time been impaired, and points especially to some affection of the heart

manifesting itself in changes in colour, breathlessness, and acute attacks of nervous depression.” One evening, when Sir Charles was taking his customary walk between the rows of yew trees, he saw the awesome hound; he became greatly frightened and ran as fast as he could toward Baskerville Hall. He fell dead before he reached the house. In this instance, the intense mental excitement and the tremendous physical effort caused heart failure—perhaps ventricular failure produced by a diseased myocardium rather than a coronary occlusion.

Physical exertion does not necessarily produce a coronary occlusion. It was once thought that patients suffering from angina pectoris, or who actually had coronary disease, should not exert themselves physically. They were warned, for example, not to run to catch a streetcar. A more liberal view is held today; most patients are urged to exercise themselves. In point of fact, many coronary occlusions occur when the patient is at complete rest or performing some slight task.

An interesting allusion to cardiac neurosis may be found in *The Sign of the Four*. When Dr. Watson was introduced to Thaddeus Sholto, the latter became greatly excited and said: “A doctor, eh?... Have you your stethoscope? Might I ask you—would you have the kindness? I have grave doubts as to my mitral valve, if you would be so very good. The aortic I may rely upon, but I would value your opinion upon the mitral.” After Dr. Watson had listened to Sholto’s heart, and had assured him that he had no cause for uneasiness, the patient turned to Miss Morstan and said: “You will excuse my anxiety, Miss Morstan ... I am a great sufferer, and I have long had suspicions as to that valve. I am delighted to hear that they are unwarranted.”

The above clearly depicts a man who is unduly, and probably needlessly, alarmed about the condition of his heart; the diagnosis of cardiac neurosis can obviously be made. This could have been brought about by some physician who, in an ill-advised moment, had hinted to the patient that he might be suffering from a cardiac disorder. Such a condition is spoken of as “iatrogenic heart disease.” Unfortunately, a good many people suffer from cardioneurosis, and the majority probably would be benefited by consulting a psychiatrist. The doctor might be able to ascertain the underlying cause for the patient’s fear.

These passages from *The Sign of the Four* show that Dr. Watson was acquainted with valvular heart disease; they are, however, rather farfetched, if not ludicrous, and probably could not be appreciated by the nonmedical reader. Watson does not often attempt to show his erudition or weary his readers by employing technical terms. A good example of this matter-of-factness may be found in *The Crooked Man*: “The injury from which this unfortunate veteran was suffering was found to be a jagged cut some two inches long at the back part of his head, which had evidently been caused by a violent blow from a blunt weapon.” The above is written in a language that anyone can understand.

In the novel *A Study in Scarlet*, may be found an interesting reference to an aortic aneurism. Jefferson Hope told Dr. Watson to put his hand over his heart. Watson writes:

I did so, and became at once conscious of an extraordinary throbbing and commotion which was going on inside. The walls of the chest seemed to thrill and quiver as a frail building would do inside when some powerful engine was at work. In the silence of the room I could hear a dull humming and buzzing noise which proceeded from the same source.

Dr. Watson promptly diagnosed it as an aortic aneurism. Just before Watson made his examination, Jefferson Hope had put up a terrific struggle to escape capture, and it had taken four men to subdue him:

So powerful and fierce was he that the four of us were shaken off again and again. He appeared to have the convulsive strength of a man in an epileptic fit. It was not until Lestrade succeeded in getting his hand inside his neckcloth and half strangling him that we made him realize that his struggles were of no avail.

The remarkable thing is that this man was suffering from an aortic aneurism which was on the point of bursting. Why the aneurism did not burst during the terrific struggle will always be a mystery. We find later in the story that it subsequently did burst.

Several references to the pulse may be found in the tales. When the pompous if dignified Thorney Huxtable, M.A., Ph.D., collapsed in Holmes’

living room, Dr. Watson attended him, and described the condition as one of severe exhaustion; he spoke of the pulse as being thready in character (*The Adventure of the Priory School*). In *The Adventure of the Stock Broker's Clerk*, a man had tried to hang himself, but Dr. Watson helped rescue him just in time. The patient was placed on the floor, and the good doctor immediately felt his pulse; he reported that it was feeble and intermittent.

At the time Dr. Watson practiced medicine, more attention was probably paid to the study of the pulse than now obtains. The experienced physician knows that considerable information can be obtained by studying the pulse. Not only can the rate be ascertained, but the character of the beat as well: whether it is full or bounding, or thready or feeble, and the like. By digital pressure on the radial artery, a rough idea can be obtained of the systemic blood pressure, and at the same time some information can be gained as to whether the artery shows any sclerosis.

Some physicians of the older school feel that the younger generation of physicians neglects the pulse, and its study is becoming a lost art. The more recently trained physicians are more apt to depend upon certain instruments of precision. Be that as it may, we have already seen that Dr. Watson attempted to gain as much information as he could by feeling the pulse.

On one occasion (*The Adventure of Shoscombe Old Place*), Dr. Watson speaks of a failing heart; he describes a patient who had a dropsical condition. When reading about a failing heart in English medical literature, one cannot but think of William Withering (1741-99), who was the first physician to employ digitalis therapeutically as a single pharmacologic agent. He reported his findings many years before Dr. Watson practiced medicine. It is singular that Dr. Watson never mentioned any digitalis preparation in the Sherlock Holmes stories. He does mention amyl nitrite, but not in connection with any heart condition; he also mentions morphine as a drug to assuage pain, but again not in connection with the heart.

Since the time Dr. Watson practiced medicine in London, great strides have been made in the diagnosis and treatment of cardiovascular disorders. The electrocardiogram and other instruments have been developed for diagnosing disorders of the heart and vessels. Methods of treatment, such as the use of antibiotic drugs, and agents for reducing high blood pressure, have been of inestimable value.

In this country during the past few years, the citizens have become increasingly aware of the value of research in cardiovascular disease. It is noteworthy that a part of this interest was stimulated by the fact that President Dwight D. Eisenhower suffered a coronary occlusion and later a mild cerebral accident.

In conclusion, when Dr. Watson practiced medicine, he did not have any effective drugs at his disposal to combat hypertension, or any drugs, such as dicumarol, to lessen the danger of further coronary occlusions. We may be assured, however, that those agents he did have were used to the best of his ability, and he did all he could to restore the patient to health. No man could do more.

THE PHYSIOLOGIC DOCTOR WATSON

“Surely your medical experience will tell you, Watson, that weakness in one limb is often compensated for by exceptional strength in the other.”

The Man with the Twisted Lip

Since I am a professional physiologist, references made to physiologic matters in the tales are of especial interest to me. Physiology, as the term is now used, may be defined as the science which deals with the function of an organism and its parts, in contrast with anatomy, which is concerned only with the structure of an organism.

Dr. Watson took his work at the University of London Medical School, and was graduated in 1878—eighty years ago. We may assume that he received good training in physiology; at any rate, he apparently had a healthy respect for this important science. Let us examine the references he makes to physiologic matters in the tales.

Physiologists

In *The Adventure of the Creeping Man*, a physiologist, Professor Presbury, is the central figure. Dr. Watson speaks of him as a renowned “Camford” physiologist. The word “Camford” is a happy choice. It strikes one that it could be a combination of two words, Cambridge and Oxford. Whether Dr. Watson had these two famous schools in mind, I cannot say, but I am inclined to believe he did. It is noteworthy that he depicts Professor Presbury as a wealthy man. On one occasion, Sherlock Holmes and Doctor Watson called at Presbury’s home. The good doctor wrote glowingly about the physiologist’s charming house and beautiful grounds, and emphasized that the professor was surrounded with every sign of luxury.

This is truly an unusual picture, for not many professional physiologists are wealthy, and they certainly are not surrounded with luxury. Professor

Presbury, fortunately, was an exception. He must have been a wealthy man in his own right, or have married a woman of means, for surely he did not become wealthy by teaching physiology.

Professor Presbury's assistant, Mr. Trevor Bennett, is interestingly pictured as a handsome, tall young man, and elegantly dressed. He was portrayed as having the shyness of the student rather than the self-possession of a man of the world. We can readily understand his shyness and his studious manner, but it is more difficult to imagine his elegant appearance. Bennett, we find later, had a medical degree, so perhaps he enjoyed a consulting clinical practice, which might account for his exceptional elegance.

Mr. Bennett's careful grooming is in contrast to the usual situation, because physiologists, I fear, do not have the reputation for being well-dressed people. They are apt to be somewhat indifferent about their personal appearance, especially when at work in the laboratory. This is quite understandable, since they often work with such animals as cats, dogs, and monkeys, and with even larger animals like sheep, goats, and pigs. Some of the more fastidious scientists don old clothes for such occasions, but this is not always convenient. At any rate, the average physiologist is not known for his elegant appearance—and for good measure, I will throw in the pharmacologists also! Several living examples of each could be cited to prove my point, but I must refrain from embarrassing my distinguished friends and colleagues. At least Mr. Bennett may be considered an exception.

Endocrinology

In the story discussed above, the author uses the theme of endocrinology. Professor Presbury, widower in his early sixties, had fallen madly in love with a very young girl. In his desire to regain his lost youth, the professor administered to himself injections of serum prepared from apes. A conniving Prague scientist had prepared the material from the langur, which Watson dramatically described as one of the higher apes, distinguished by a black face, and an inhabitant of the slopes of the Himalayas.

According to the story, the serum had a profound effect on the professor. Following the serum injections, which he took at nine-day intervals, he acquired the characteristics of an ape, not only by simulating the mode of locomotion of this animal but by developing, also, an uncanny ability to climb.

One night, following a serum injection, the professor got into grave difficulty when his heretofore faithful wolfhound Roy, which he was bedeviling, attacked him viciously. As Sherlock Holmes told Dr. Watson, the wolfhound thought he was attacking the monkey, not his master. Presbury was saved by the quick action of Holmes and Watson.

The plot is fantastic, but interestingly told. Today this story would be regarded as science-fiction. It was written at a time when the subject of rejuvenescence had been made popular by the endocrine studies of Steinach and of Voronoff, the European scientists. The father of endocrinological studies was, of course, Charles-Edouard Brown-Séquard, who had reported his studies on testicular extracts in 1889. Dr. Watson was undoubtedly familiar with Brown-Séquard's work. In *The Adventure of the Creeping Man*, Holmes points out a moral when he remarks to Watson that, if one leaves the straight road to destiny, even the highest type man may revert to the animal.

Digestion

Let us turn to the physiology of digestion. Dr. Watson obviously felt that one should eat sparingly if brain work is to be done. We find him saying that his friend Holmes had eaten no breakfast, because in his more intense moments he allowed himself no food. Holmes contended that he did not have energy to spare for both nerve force and digestion (*The Adventure of the Norwood Builder*). On another occasion, Holmes, when talking to Doctor Watson, emphasized the fact that starvation refines the faculties. He insisted that during digestion the brain is robbed of blood, because blood is needed in the splanchnic area. Holmes insisted that he himself was a brain, and the rest of his body a mere appendix (*The Adventure of the Mazarin Stone*).

The ideas expressed in the above paragraph hold our physiologic interest because, at the time Watson wrote, the concept he expressed was

the accepted one. Later work has shown that this theory is not tenable. The modern view is that digestion does not influence the blood supply to the brain. In point of fact, nature has taken careful precautions to ensure the brain of a rich blood supply at all times under normal physiologic conditions.

Although the brain does not suffer in its blood supply during digestion, it is well known that the body is capable of shunting blood to those regions which are functioning and are, therefore, in need of it. If a hearty meal is eaten, a rich supply of blood is required to digest and absorb it. Athletes know that a large meal should not be eaten before taking strenuous exercise. Indeed, if a substantial amount of food be eaten preceding intense physical exertion, the athlete may become nauseated or even vomit.

The voluntary muscles need blood when functioning, and the greater the amount of exercise, the more blood is needed. The circulatory system cannot supply both the splanchnic region and the skeletal muscles under these conditions, so some part of the body must suffer. The nausea and vomiting are, in a measure at least, produced by anemia of the stomach. In the example just given, there is also a nervous component, which in some individuals plays an extremely important part.

Let us return to the blood supply of the brain. There is some evidence that the brain needs a slightly increased blood supply during lively mental activity, but not all physiologists will accept these findings. However, this may safely be stated: the metabolism of the brain is high at all times and is not affected by mental work. If one indulges in "heavy thinking," one simply directs the activities of the nerve cells in the brain, which already are active.

If we accept the hypothesis that increased mental activity does call for slightly more blood to the brain, then the mind actually may be more keen and alert when the circulatory system does not have to sustain the burden of digestion and absorption of food. The fact that there is often a feeling of well-being following a satisfying meal does not negate this concept. There are those who maintain that the mind functions best during a state of moderate fasting. Dr. Watson evidently believed this, as has been previously stated. All physiologists would not agree with this concept. Undoubtedly,

there is a wide individual variation in this regard, and it would be unwise to make a sweeping generalization.

A Weaker Sex?

Dr. Watson implies, in *The Adventure of the Devil's Foot*, that physiologically the female has a weaker constitution than the male. In this story, it will be recalled that two men and one woman were exposed to the fumes of the powdered devil's-foot root. Watson stated that only the woman was killed, presumably because she was the more sensitive. This is indeed a challenging statement.

Considerable proof can be adduced to show that the female is hardier than the male. The old cliché, "the weaker sex," can be seriously questioned. It is well known, for example, that women can withstand a high blood pressure for years. In some instances, although by no means all, it seems to do them but little harm, since in spite of it they may live to a ripe old age. Hypertension in the male is generally a serious condition. Many men die comparatively young from heart failure or from cerebral hemorrhage due to hypertensive states.

Recently, it has been shown that female animals, such as rats, can withstand acute oxygen-want far better than males. I do not wish to belabor the point that the female may be hardier than the male, but two more bits of evidence can be presented. Data compiled by life-insurance companies definitely show that there are more widows in this country than widowers. Lastly, more boy babies die than girl babies.

These arguments seem to prove convincingly that the male and not the female is the weaker sex. The problem, however, is not as simple as this, but space does not permit further critical treatment of the subject. In fairness to Dr. Watson, it must be stated that many of the facts just presented were not known when he wrote his entertaining but gruesome story about the devil's-foot root.

Mental Development in the Young

We find, in *The Adventure of the Sussex Vampire*, an interesting and significant reference to mental development in the young. Holmes, speaking of a child, remarked in this story that the child probably had a very well-developed mind, because his body had been circumscribed in action. Obviously, a child who has a definite physical handicap cannot romp and play like other healthy, vigorous children. He is forced to find other outlets, and as a result probably reads a great deal, and is likely, moreover, to be thrown in with older people. As a consequence, his mind presumably is more precociously trained. There are many physically handicapped children, and this entire matter deserves more critical psychologic study than it has so far received.

Curare and Physiology

Dr. Watson was acquainted, at least to a degree, with the action of curare. Actually, he used it as a lethal agent in two of his tales: in his novel *A Study in Scarlet*, and in his story, *The Adventure of the Sussex Vampire*. In the former, he was careless of the way he handled curare in his plot, for he ascribed actions to this drug which it simply does not possess. But in the latter instance, he uses curare in a masterly manner. He points out, for example, that if a child were pricked with an arrow which had been dipped in a solution of curare, death might ensue by muscular paralysis if the curare were allowed to be absorbed. Incidentally, this story can be highly recommended not only to physiologists and pharmacologists but to all medical and biological students as well.

It is likely that Watson became familiar with curare in his work in the physiology laboratory, for about twenty years before he enrolled in medical school, Claude Bernard, the famous French physiologist, had published his studies on curare.

Muscle Physiology

Let us now consider references to muscle physiology. Both Sherlock Holmes and Dr. Watson are portrayed as splendid athletes. In *The Hound of the Baskervilles*, Watson writes about their heroic efforts to save Sir Henry from the awesome hound. He states that he had never seen Holmes run as

fast as he did that night. He remarks further that he (Watson) was reckoned fleet of foot, but Holmes easily outpaced him. In physiologic parlance, the muscles of Holmes and Watson presumably had short reflex times. Some men are born fast runners. The exact reason for this is not fully understood. Probably the entire neuromuscular apparatus, including the higher nervous centers, is implicated. Another factor is the mechanical advantage involved in the muscle-bone relationship. All these factors undoubtedly have to do with heredity. The entire picture is complicated. We do know, however, that training alone cannot produce champions.

In another instance, Holmes discusses with Watson the footprints found in the yew alley, which were presumably made by Sir Charles Baskerville. It looked as if he had walked on tiptoe for at least a part of the way. Watson asked why Sir Charles had walked on tiptoe. Holmes instantly replied: "He was running, Watson—running desperately, running for his life, running until he burst his heart and fell dead upon his face."

On another occasion, Holmes described the gait of a man: "... he was running hard with occasional little springs, such as a weary man gives who is little accustomed to set any tax upon his legs" (*The Adventure of the Beryl Coronet*). On still another occasion, we find an interesting reference to muscle physiology—one which only recently has been scientifically demonstrated. In *The Man with a Twisted Lip* Holmes makes this pregnant remark: "He is a cripple in the sense that he walks with a limp; but in other respects he appears to be a powerful and well-nurtured man. Surely your medical experience will tell you, Watson, that weakness in one limb is often compensated for by exceptional strength in the other."

Muscle physiology is a highly specialized field. Considerable information is known about the chemical changes which take place during muscular contractions, but much more needs to be learned. One of the factors of supreme interest is the short period of time required for these complicated chemical and heat reactions. The problem is of more than academic interest, for it must be remembered that, in spite of all the modern machines at man's disposal, the greater part of the work in this world is still done by muscular activity.

Any light thrown on the exact mechanism of muscle contraction may eventually prove to be of extreme practical benefit. It is indeed difficult to

overemphasize this point. Finally, there are certain serious disorders of muscular functioning which may not only incapacitate individuals but actually cause their death. The distressing condition of patients suffering from muscular dystrophy is a poignant example. For this reason, if no other, a complete understanding of muscular contraction is of paramount importance.

Since the time when Dr. Watson studied physiology—eighty years ago—there have been vast developments in this field of science. There is hardly an area of scientific endeavor in which more research has been done or is in progress.

SHERLOCK HOLMES AND DOCTOR WATSON, PERENNIAL ATHLETES

“Sherlock Holmes was a man who seldom took exercise for exercise’s sake ... he looked upon aimless bodily exertion as a waste of energy.”

The Adventure of the Beryl Coronet

Sherlock Holmes and Dr. Watson were unquestionably men of outstanding athletic ability. Holmes was a single-stick player, a boxer, and a swordsman (*A Study in Scarlet*). Watson had played rugby in college, and probably had engaged in other collegiate sports. They were strong men, quick of action and fleet of foot.

Quickness of action surely was needed to save Sir Henry Baskerville from the hound. It will be recalled that the fog from the Grimpen Mire that autumn night had spread over the path which Sir Henry was to follow from Stapleton’s house to Baskerville Hall. As a consequence, the visibility was so low that Sherlock Holmes and his companions could not see the hound until it was dangerously close to its intended victim. Watson describes the episode dramatically: “Never have I seen a man run as Holmes ran that night. I am reckoned fleet of foot, but he outpaced me as much as I outpaced the little professional” (*The Hound of the Baskervilles*). At any rate, we know that these determined men, led by Holmes, reached Sir Henry just in time and killed the hound before it could harm him.

Watson’s statement, “Never have I seen a man run as Holmes ran that night,” bespeaks high praise of Holmes as an exceptionally fast man. Watson, when a college student, had doubtless seen first-class trackmen perform, but nevertheless he distinctly implies that Holmes as a runner was in a class by himself. Holmes was making an effort to save a man’s life, but so were Watson and Lestrade, the Scotland Yard detective. Holmes, it is true, had the build of a trackman, since he was tall and lithe, whereas Watson had a decidedly burly physique, and Lestrade is described as rather

short in stature. Watson, despite his stocky build, admits that he himself was a sprinter of parts, but he was obviously no match for Holmes.

Not only was Holmes a fast runner, but a skilled boxer as well. Watson writes: “He was undoubtedly one of the finest boxers of his weight I have seen ...” (*The Yellow Face*). A real compliment, for Watson was a keen sportsman who probably had had occasion to attend many boxing matches. Holmes was six feet tall—at least he so described himself (*The Adventure of the Three Students*). The suggestion has been made that he weighed about 11 stone (154 pounds) and had a physique resembling that of Bob Fitzsimmons,^[5] that is, he was rangy. A boxer with a long reach has, of course, a decided advantage. Holmes, too, was fast on his feet, as previously mentioned. Added to these attributes, he had a superior intelligence, which enabled him to analyze his opponent’s tactics and to outsmart him. He doubtless was a dangerous man in the ring. There is evidence that this is so. He gave an extremely good account of himself when he tackled such rough customers as Roaring Jack Woodley (*The Adventure of the Solitary Cyclist*), Joseph Harris (*The Naval Treaty*), and an anonymous ruffian (*The Final Problem*); furthermore, the professional boxer McMurdo (*The Sign of the Four*) had a high regard for Holmes’ boxing ability.

Besides being a runner and a boxer, Holmes was a wrestler of no mean ability. He tangled with such villains as Colonel Sebastian Moran, the second most dangerous man in London (*The Adventure of the Empty House*); Professor Moriarty, the Napoleon of crime (*The Final Problem*); and the vigorous and powerful German master spy Von Borck (*The Last Bow*). It has been suggested,^[6] and I believe rightly, that Holmes used “baritsu” when dealing with these dangerous characters. They were real killers and would have stopped at nothing. Sherlock Holmes could not afford to take chances when he encountered them. In many instances, Holmes had the dependable and powerful Watson at his side; but whether alone or in company, we know he nearly always got his man.

We infer, especially from a certain episode, that Holmes possessed muscles of steel and had an iron grip. It will be remembered that on one occasion the fierce and gigantic Dr. Grimesby Roylott strode unannounced into the chambers at 221B Baker Street. In a fit of temper, he seized a poker, “... and bent it into a curve” (*The Adventure of the Speckled Band*).

After the unpleasant visitor had left, Holmes without much apparent effort straightened the poker out—a task requiring more strength than the initial bending.

I have always felt that, in spite of this startling demonstration on the part of Holmes, for sheer bull-like strength Dr. Watson undoubtedly excelled him. It is true that once at least Watson met his match. He relates that on one occasion he was assaulted by an extremely powerful and active man who sprang upon him and nearly choked him to death (*The Disappearance of Lady Frances Carfax*). This is an unusual incident, and Watson must have been unprepared, for in a rough-and-tumble fight he could take care of himself in fast company. He had the courage and tenacity of an English bulldog, and, what is more important, knew how to use his strength. Watson did not pretend to be as quick on his feet as Holmes, but he had the other qualities just mentioned to a superlative degree, and these count for a great deal in a catch-as-catch-can bout. Holmes was cognizant of Watson's prowess, for once the great detective stated that when a man was in a tight place there was no better man to have on one's side than Watson.

An interesting query can be posed: How did these men keep themselves in reasonably sound physical condition? Watson does not help us to answer this question satisfactorily, for he has this to say about Holmes:

Sherlock Holmes was a man who seldom took exercise for exercise's sake. Few men were capable of greater muscular effort, and he was undoubtedly one of the finest boxers of his weight that I have ever seen; but he looked upon aimless bodily exertion as a waste of energy, and he seldom bestirred himself save where there was some professional object to be served. Then he was absolutely untiring and indefatigable. That he should have kept himself in training under such circumstances is remarkable, but his diet was usually of the sparest, and his habits were simple to the verge of austerity.

The Adventure of the Beryl Coronet

Watson makes two statements which especially require critical comment: the first one, "Few men were capable of greater muscular effort ..." and the second, "Then he was absolutely untiring and indefatigable."

These statements are not convincing, because they are not consistent. Watson plainly indicates that Holmes did not believe in training; and when the record is examined, we discover that actually both Holmes and Watson led sedentary lives. Neither of them engaged in any sport while residing at 221B Baker Street. The only exercise they took, when not working on a case, was walking.

It might be in order to digress a moment and speak of the physiology of exercise. As far as physical training is concerned, we know that there is no substitute for strenuous exercise. The physiology of training is not yet well understood, and but few objective criteria are recognized. There is one criterion, however, which is widely known, and that is the muscular hypertrophy produced by exercise. The classical example is the tremendously developed muscles of the arm of the blacksmith. Most of us in childhood learned Longfellow's apt lines: "The smith, a mighty man is he, with large and sinewy hands; And the muscles of his brawny arms are strong as iron bands."

Besides muscular development, there are other factors, although less objective in character and less clearly understood, which are brought about by training. Among these are adaptive changes in the cardiovascular and respiratory system, and certain chemical changes in the blood.

Several examples can be given to show that Dr. Watson appreciated the physiologic symptoms brought about by lack of physical training. On one occasion, he describes the unexpected visit to 221B Baker Street, upon a winter's morning, of Mr. Alexander Holder, the financier of Threadneedle Street. "He was a man of about fifty, tall, portly, and imposing ... he was running hard, with occasional little springs, such as a weary man gives who is little accustomed to set any tax upon his legs." Watson continues: "... the man, puffing and blowing, rushed at our door ..." (*The Adventure of the Beryl Coronet*). This excited and distracted individual finally composed himself and apologetically explained that, instead of taking a cab (because cabs go slowly through the snow), he had hurried to Baker Street on foot. Said he: "That is why I was so out of breath, for I am a man who takes very little exercise." In this connection, one is reminded of Lewis Carroll's lines:

"But wait a bit," the Oysters cried,

“Before we have our chat;
For some of us are out of breath,
And all of us are fat.”

On another occasion, Watson discloses that, because of his sedentary life, he became “blown” and was forced to slow down, whereas Holmes never faltered, because he was always in training. Watson admitted that his sedentary life had begun to tell upon him and he had to fall behind. Holmes, on the other hand, never slackened his pace, and apparently had an inexhaustible supply of nervous energy (*The Adventure of the Solitary Cyclist*).

Another example is found in *The Hound of the Baskervilles*. It will be remembered that, on a damp, sullen evening, Dr. Watson and Sir Henry attempted to catch the escaped criminal Selden, who was hiding out on the moor. Watson, in his report to Holmes, writes:

We were both swift runners and in fairly good training, but we soon found that we had no chance of overtaking him.... We ran and ran until we were completely blown, but the space between us grew ever wider. Finally we stopped and sat panting on two rocks....

In this instance, Watson mentions that he was in fairly good physical condition, although he does not tell us how he managed to reach that state. We must assume that he had been keeping his weight down by dieting and probably taking long walks on the lonely moor. These two factors alone, however, are not sufficient to fit a man for extraordinary and prolonged physical exertion. As pointed out previously, the only way this can be done is by daily, well-regulated, arduous exercise.

In the matter of physical fitness, there is another side of the shield which we must examine. A few individuals, although not many, in spite of their sedentary habits, are apparently at all times physically fit. The muscles of these men seem to keep their firmness and power even though they are but little used. This is difficult to understand. Sherlock Holmes probably was one of these unusual people.

I doubt whether Watson would fit into this category. The good doctor probably had a tendency to put on weight rather easily, and one gets the

impression from reading the tales that Watson liked his three square meals each day. He was, to my mind, a typical beefeater. Furthermore, he smoked a great deal, but for that matter so did Holmes. Smoking is thought to interfere with physical fitness; it is said to be “hard on the wind.” Athletes in training are invariably forbidden to smoke. Just how smoking affects physical fitness is not known. It is generally conceded, however, that tobacco in any form, especially if used to excess, reduces both mental and physical efficiency. There is considerable scientific proof for this statement, although this is not the place to present it.

The beneficial effect of even a moderate amount of exercise has been questioned by otherwise intelligent men. A brilliant colleague of mine, for example, contends that the only thing exercise does is to make one fit for still more exercise *ad infinitum*. Well, in a measure this is true, for exercise really is the basis of training, as has been repeatedly emphasized. One famous educator in this country has publicly stated that, when he feels like exercising, he lies down until the feeling passes away. Then there is the story of Chauncey Depew, who lived to the age of ninety-four. When asked what exercise he took to maintain his health, he replied: “The only exercise I have ever taken is to serve as pallbearer for my friends who exercised.” I suspect that these members of the intelligentsia, in their zeal to make their points, have purposely overstated their case.

Be that as it may, many people insist that a certain amount of exercise causes them to feel better. Admittedly, it is difficult for them to explain just what is meant when they say “feel better”—it is a subjective rather than an objective matter. Several reasons can be mentioned, however, why moderate exercise seems beneficial to many people.

To those whose work keeps them constantly at a desk, exercise comes as a welcome relief from monotony. Physiologically, we know that a change of occupation in many instances is as good as a rest. A brisk walk through the park—or better still, in the country—brings to the eye a shifting panorama and relieves boredom and fatigue. Many people enjoy sports, such as a leisurely swim or a short round of golf. The pursuit of these not only calls for exercise, but rests the mind from vexing problems. In fact, there is something intangible about it all; in many individuals, a pronounced

psychic component unquestionably plays an important part. Keeping fit, for example, often produces a mental lift and makes some people feel quite content with themselves. This feeling may assume sizable, if not untoward, proportions. The person who takes exercise regularly, who keeps his waistline down, and maintains himself in a fairly good physical state, may feel smug and distinctly superior. He often looks down with pity at his friends who do not believe “in exercise for exercise’s sake,” and, we might add, whose figures confirm and illustrate the fact.

In discussing the philosophy of exercise, we wish to make clear that there is no proof that strenuous muscular activity is beneficial to the body. Indeed, great physical exertion in those past middle age may be deleterious to the organism. The late distinguished English physiologist, Professor F. A. Bainbridge, has happily stated: “The exercise consequent upon the ordinary activities of life is sufficient.”

Exercise, however, should not be counted upon to bring about weight reduction. It has been shown, for example, working with marathon runners at the Harvard Fatigue Laboratories, that it took two and one-half hours of steady running to burn up two pounds of carbohydrates. It has been facetiously pointed out that the most telling exercise leading to a trim figure is the simple procedure of pushing the chair away from the dining table.

The current belief is that people who keep reasonably fit appear healthier than those who do not. Watson caught this thought when he wrote that Mr. Garrideb had a cadaverous face, his skin was dull, and it resembled that of a person who never exercised (*The Adventure of the Three Garridebs*).

In passing, we find a reference concerning a means of taking exercise which shows the austere Holmes in a light frame of mind. He twits Watson about an individual’s using only one dumbbell, and facetiously states that such a person would not develop his muscles symmetrically, and as a consequence might suffer from curvature of the spine (*The Valley of Fear*). Holmes does not tell the whole story. The single dumbbell which worried him could have been used alternately by the right and left hand, and thus unilateral development would be prevented. Watson, as a medical man, knew this; but like a good soldier he let Holmes have his fun and made no reply.

In conclusion, we may say that the principal question we have raised still remains unanswered: that is, how did Holmes and Watson keep themselves in good physical condition? This will always be a mystery. Search as we will, we can find no evidence that either Holmes or Watson kept himself in training. This seems remarkable since there were times when they needed strength and stamina in order to put their foes out of commission, and indeed there were occasions when their very lives depended upon it.

It is difficult to believe that unless Holmes and Watson kept themselves in fairly good condition they could have held their own against strong and desperate men, who presumably were in good fighting form. When one considers how important it was for our heroes to keep physically fit, one is drawn to the irresistible conclusion that they actually took much more exercise and kept themselves in far better training than appears in the tales. Besides their long walks through London, they probably took setting-up exercises at 221B Baker Street. They may even have done some weight lifting or have used dumbbells (since Holmes was so concerned about finding a single dumbbell); perhaps they boxed with each other. It is even possible that they did some swimming. It will be recalled that, after his retirement, Holmes frequently went swimming (*The Adventure of the Lion's Mane*). They may have engaged in still other forms of exercise. I, for one, am ready to believe that they did not lead such sedentary lives as Dr. Watson would have us think.

THE THERAPEUTIC DOCTOR WATSON

“I had obtained good results in such cases by the inhalation of nitrite of amyl....”

The Resident Patient

John H. Watson, M.D., practiced medicine in the city of London in the early eighties of this past century. This was a long time ago—nearly seventy-five years. At the time of which we write, Roentgen had not as yet discovered the X-ray, and Ehrlich’s “606” (salvarsan) was not available for the treatment of syphilis. Banting and Best, the Canadian physiologists, had not given insulin to the world, nor had Minot and Murphy of Harvard discovered that liver extract was an efficient agent in the treatment of pernicious anemia, a heretofore fatal blood disease. The sulfonamide drugs and the antibiotics, such as penicillin and streptomycin, presumably were only a dream.

From what has been related, it appears that Dr. Watson’s therapeutic armamentarium, insofar as efficacious drugs are concerned, was sadly limited. But the fact must not be forgotten that he had at his command a number of powerful medicines: morphine, cocaine, belladonna, and strychnine, to name a few.

Let us turn to the stories and see what medicines Dr. Watson used in his practice and under what conditions they were employed.

It appears that iodoform and silver nitrate were well known drugs, for Holmes on one occasion remarked to him: “... if a gentleman walks into my rooms smelling of iodoform, with a black mark of nitrate of silver upon his right forefinger.... I would be dull indeed if I do not pronounce him to be an active member of the medical profession” (*A Scandal in Bohemia*).

Iodoform was introduced as an antiseptic about the year 1879. For many years it was widely employed as a dusting powder, especially to diminish infection of open wounds and to promote granulation. Its persistent and penetrating odor, as well as its relative ineffectiveness, gradually brought it

into disfavor. This is probably just as well, for its use is not entirely without danger, especially when iodoform is applied over an extensive area where a good deal of it can be absorbed.

Silver nitrate still enjoys a wide usage. It is employed to produce astringent, antiseptic, and caustic effects. Even many laymen know that silver nitrate will remove “proud flesh.” Unfortunately, it has the property of staining the skin “with a black mark” (as Holmes remarked), which incidentally is often difficult to remove. It may persist until it is virtually worn off, although recent and superficial stains can be removed by painting the area with a ten per cent solution of potassium iodide.

As might be expected, Dr. Watson found it necessary on several occasions to resort to morphine to alleviate pain. We find, for example, that when Kitty Winter, the cast-off mistress of the infamous Baron Gruner, threw vitriol in the Baron’s face, Dr. Watson administered a hypodermic of morphine to the wretched Baron to ease the intense suffering (*The Adventure of the Illustrious Client*). On another occasion, when Professor Presbury was savagely attacked by his wolfhound, Watson immediately gave morphine to the unfortunate professor (*The Adventure of the Creeping Man*). In still another instance, when Carlos, the mastiff, had attacked Rucastle, Watson probably would have used morphine had he had it in his possession. He wrote: “I did what I could to relieve the pain” (*The Adventure of the Copper Beeches*). He did not state, however, what agent he employed.

On one occasion, at least, it was necessary to administer morphine to Sherlock Holmes. He had been attacked and beaten severely by two ruffians. Sir Leslie Oakshott, the eminent London surgeon, had been called in on the case. He later tersely remarked to Watson that Holmes suffered from lacerated scalp wounds, and there were also some bruises. It had been necessary to take several stitches. Morphine had been administered (*The Adventure of the Illustrious Client*). Morphine, because it is a dangerous habit-forming drug, should be used only when absolutely indicated. It is noteworthy that Dr. Watson administered morphine only when the patient had suffered a grave physical injury. He is to be commended for his conservative attitude.

On many occasions, when assisting Sherlock Holmes with his cases, Dr. Watson found it necessary to administer a stimulant to those individuals who had suffered a physical injury or had been under great emotional stress. His favorite remedy for such conditions was brandy. Numerous instances may be cited when this agent was employed as an emergency measure. Dr. Watson obviously believed firmly in the efficacy of brandy as a stimulant, and, indeed, claimed to have saved the life of an individual by its use:

Mr. Melas, however, still lived, and in less than an hour, with the aid of ammonia and brandy I had the satisfaction of seeing him open his eyes, and of knowing that my hand had drawn him back from that dark valley in which all paths meet.

The Greek Interpreter

Dr. Watson in this beautiful passage expresses the wish of every physician.

Brandy and whiskey were not the only stimulants used by Dr. Watson; in point of fact, he employed several others. We have noted previously that ammonia was administered along with brandy to Mr. Melas, the Greek interpreter. Physicians will recall that ammonia is the principal ingredient in so-called smelling salts. The vapors of ammonia when inhaled stimulate the medullary centers, causing an increase in respiration, and by action on the vasomotor center produce a rise in blood pressure.

In his heroic efforts to save the life of a woman who had been chloroformed nearly to death, Dr. Watson gave ether intravenously with successful results (*The Disappearance of Lady Frances Carfax*). At any rate, he records that the patient regained consciousness. Ether is not often given in this manner as a stimulant, and the question could be raised whether it was indicated under the condition which faced Dr. Watson. I am afraid that this was an instance in which the patient lived in spite of the treatment.

A popular and effective stimulant, namely, caffeine, was used to bring a young woman out of a stuporous state produced by opium. Dr. Watson relates that a couple of cups of very strong coffee brought her out of this state (*The Adventure of Wisteria Lodge*). Dr. Watson used the classical agent. Strong coffee is still used today in morphine poisoning, although

relatively new preparations, such as metrazol, nikethamide, and amphetamine are at present the drugs of choice. Coffee is, of course, easy to obtain, is a relatively harmless stimulant, and has the pleasing virtue of not producing disagreeable aftereffects.

On one occasion, the nature of the stimulant administered was not disclosed. We find the following: "... while his nurse poured him out a glass of some stimulating medicine" (*The Naval Treaty*). We can only guess at the contents of the glass, but more than likely it contained either brandy or whiskey. We have previously pointed out that these agents were frequently prescribed as stimulants.

Dr. Watson, strangely enough, administered amyl nitrite to a patient who presumably was suffering from catalepsy. He writes, "I had obtained good results in such cases by the inhalation of nitrite of amyl...." (*The Resident Patient*). It will be recalled that this rather rare condition is characterized by a rigidity of all the skeletal muscles; the patient, as a consequence, tends to remain in any position in which he is placed. The reason Dr. Watson used amyl nitrite in this instance is not clear. The nitrites are frequently prescribed to reduce systemic blood pressure or to relieve angina pectoris. Surely there is no evidence in the literature that the nitrites are of aid in cataleptic states. We can dismiss the matter by assuming that in this instance Dr. Watson used amyl nitrite empirically.

Oil is mentioned as a therapeutic agent in at least two of the stories. Both Dr. Watson and Sherlock Holmes had occasion for its use. When Baron Gruner had vitriol thrown in his face, Dr. Watson tried to alleviate the poor wretch's pain by bathing his face in oil and applying cotton to the raw areas (*The Adventure of the Illustrious Client*).

On another occasion, Holmes relates how he treated Murdoch, who had come into contact with the giant jellyfish. He used cotton-wool which had been soaked in salad oil; and he related that this application greatly relieved the pain (*The Adventure of the Lion's Mane*). This adventure took place in the country, and we may assume that there was little, if any, medication on hand. The patient was suffering intensely, and Holmes did what he could to assuage the pain with the material he had at his disposal. He showed real ingenuity by making use of a homely preparation. Morphine obviously was badly needed, and it is a pity that Dr. Watson was not there to administer it.

But it must be remembered that, at the time this happened, Holmes had retired to the Sussex Downs and Dr. Watson was in London. A perusal of the story shows, however, that the patient had been given liberal quantities of brandy, so perhaps he got along fairly well without morphine after all.

Dr. Watson used carbolized bandages to dress the hand of the unfortunate hydraulic engineer who had, a few hours previously, lost his thumb in an adventure which nearly cost him his life. "I sponged the wound, cleansed it, dressed it, and finally covered it over with cotton wadding and carbolized bandages" (*The Adventure of the Engineer's Thumb*). Carbolized bandages are no longer used in present-day surgery. Historically, it is of interest to record that carbolic acid became widely known as an antiseptic through its introduction by Lister in 1867.

Dr. Watson does not often mention hypnotics. We find in one of his stories that it was the custom of an invalid, Mr. Percy Phelps, to take some sleeping potion: "... you did not take your usual draught that night" (*The Naval Treaty*). We shall never know what the "usual draught" contained. The barbiturates which are so widely used today were unknown when Dr. Watson practiced medicine. Bromides were employed not only for sedation but for inducing sleep as well. The draught of which Watson spoke might have contained chloral hydrate, for it once enjoyed wide usage as a hypnotic. It is a potent and dangerous drug and, as generally known, is used by gangsters (among others) in what are inelegantly but expressively referred to as "knockout drops."

An interesting reference is found to an alterative in one of the stories (*The Disappearance of Lady Frances Carfax*). Holmes, meeting Watson in an establishment which specialized in Turkish baths, asked his friend why he insisted upon this type of therapy. The latter replied that a Turkish bath is actually an alterative in medicine. He explained further that an alterative was a cleanser of the system. An "alterative" is defined in *Webster's Unabridged Dictionary* as: "A medicine or treatment which gradually induces a change, and restores healthy functions without sensible evacuations." A modern medical dictionary states: "An obsolete term originally used for drugs said to reestablish healthy functions of the system." (*Dorland's Illustrated Medical Dictionary*. Philadelphia: W. B. Saunders Company, 1947).

Present-day medical students probably never heard of alteratives. Many middle-aged physicians, however, will remember them well, for formerly alteratives were used extensively in therapeutics, and included the following preparations: iodine, arsenic, iron, and the compounds of mercury. Nowadays, a more scientific attitude is taken toward these agents, and they are prescribed only if there is a specific need for them: for example, iron for certain anemias, mercury for syphilis, and so on. If one considers the esteem in which alteratives were held in the latter part of this past century—and, for that matter, in the early part of the present one—it is surprising that Dr. Watson did not mention alteratives more frequently.

One wishes that Dr. Watson had mentioned other drugs used in his practice. In the early part of the twentieth century, a survey in England showed that physicians considered that from thirty to thirty-five drugs were necessary to practice medicine satisfactorily. In the stories of Sherlock Holmes, the names of about a dozen occur. Dr. Watson probably mentioned only those which for the main part are familiar to the layman.

Drugs are, of course, an important part of the armamentarium of doctors, especially those engaged in general practice. From the evidence at hand, obviously Dr. Watson was not a therapeutic nihilist, but neither can he be accused of prescribing drugs loosely or in excessive doses.

The matter of overdosing with drugs calls for a brief comment. The famous poet-physician, Oliver Wendell Holmes, wrote: “If the whole *materia medica* (excepting opium and ether) as now used, could be sunk to the bottom of the sea, it would be all the better for mankind—and all the worse for the fishes.” These were harsh words; but when written, timely. The indiscriminate use of drugs was directly responsible for the genesis of the ill-conceived homeopathic system of medicine founded by Hahnemann. The followers of this school used drugs in such minute doses that they were virtually ineffectual. It is, indeed, an ill wind which blows no good, for the rise of homeopathy had a sobering influence on many physicians of the allopathic school who were using drugs lavishly, but not well. Let us not forget, however, that the prescribing of drugs plays only a part—albeit an important one—in the successful practice of medicine.

In conclusion, we may think of Dr. Watson briskly setting out to answer the call, carrying his trusty bag filled with the recognized and popular drugs

of that period. He undoubtedly prescribed these fairly liberally, and firmly believed in their curative powers. Most of Dr. Watson's patients probably got well with the aid of, or perhaps in spite of, his prescriptions. Those who were not helped by his medication were undoubtedly benefited by his kindness and his sympathetic understanding of their ailments.

DOCTOR WATSON, GENERAL PRACTITIONER

One night ... there came a ring to my bell, about the hour when a man gives his first yawn and glances at the clock. I sat up in my chair.... I groaned for I was newly back from a weary day.

The Man with the Twisted Lip

John H. Watson, M.D., was born in England in 1852. He was graduated from the University of London Medical School in 1878 (*A Study in Scarlet*), and took his internship at St. Bartholomew's Hospital.

Since England was then engaged in a war on the Afghan frontier, Dr. Watson became attached as a military surgeon to the Fifth Northumberland Fusiliers. The English were badly defeated on July 27, 1880 by a large Afghan force at Maiwand. Dr. Watson received a painful wound—a Jezail bullet hit him in the shoulder. Later he contracted typhoid fever in a base hospital at Peshawar. He was eventually discharged from the army and returned to London with a very modest disability pension.

Not long after this, he met Sherlock Holmes through a mutual acquaintance, and they obtained comfortable living quarters at 221B Baker Street. While living with Holmes, Dr. Watson, as we know, assisted him in many of his cases and served as his chronicler.

After a time, Dr. Watson married and set up as a general practitioner in London. Doubtless, many of his experiences in his practice are reflected in his writing. It is in order to consider some of the references made to the general practice of medicine in the tales.

In one of Watson's early stories, *A Scandal in Bohemia* (written about seventy years ago), we find Holmes saying to Dr. Watson:

As to your practice, if a gentleman walks into my rooms smelling of iodoform, with a black mark of nitrate of silver upon his right forefinger and a bulge on the side of his top-hat to

show where he has secreted his stethoscope, I must be dull indeed, if I do not pronounce him to be an active member of the medical profession.

As Holmes stated, if an individual smelled of iodoform, had silver nitrate stains on his fingers, and carried his stethoscope in his hat, it would not be difficult to pronounce him a medical man. Iodoform, which was once widely used, is employed but little today, and it would be poor taste for a physician making professional calls to carry its odor with him. Silver nitrate is still widely used, but most physicians make an earnest effort to keep their hands looking neat and free of stains.

Dr. Watson had an appropriate sense of the fitness of things, and appreciated the fact that a physician should comport himself with dignity both in manner and in dress. Dr. Watson writes: “The rough-and-tumble work in Afghanistan, coming on the top of a natural Bohemianism of disposition, has made me rather more lax than befits a medical man” (*The Musgrave Ritual*). In speaking of dress, Watson states: “His dress was quiet and sober—a black frock coat, dark trousers, and a touch of color about his necktie” (*The Resident Patient*).

It is hardly necessary to say that the modern physician no longer carries his stethoscope in his hat. An interesting reference is found in the story just mentioned concerning how the general practitioner carried his instruments. One evening, Holmes and Watson returned from a walk and found a brougham waiting at their door. Holmes remarked that it belonged to a general practitioner. Watson writes: “I was sufficiently conversant with Holmes’ methods to be able to follow his reasoning, and to see the nature and state of the various medical instruments in the wicker basket which hung in the lamplight inside the brougham....”

A fact not generally known in this country is that a red lamp was once the sign of the general practitioner. Reference is made to this in *The Adventure of the Six Napoleons*. Watson writes that a bust of Napoleon, owned by a Dr. Barnicott, was found broken near his red lamp.

A pleasing manner is surely a desirable trait in a physician. There are doctors whose mere presence in the sickroom makes a patient feel better. Dr. Watson is depicted as a person not only eager to help his patients but also as a kind and sympathetic individual—attributes that are most

commendable in all physicians. Once when Dr. Watson was speaking of a lady who was emotionally upset, he writes: "We soothed and comforted her by such words as we could find." Somewhat later, he states: "I am an old campaigner.... If I can be of any assistance, either to you or my friend here, I should be indeed happy" (*The Man with the Twisted Lip*). In *The Adventure of the Sussex Vampire*, Dr. Watson again states that he would be glad to be of any service. Later, when he was asked to see a lady who had suffered a great nervous shock, he explains how he stepped up to the bed on which she was lying and spoke a few reassuring words to her, while he took her pulse and temperature.

Dr. Watson, like many practitioners, presumably had patients whom he felt he was not helping, and who were taking an inordinate amount of his time and energy. It is likely that he persuaded them to seek the aid of some other physician. Holmes probably sensed this, for on one occasion Dr. Watson asked whether a certain individual was Holmes' client. The detective remarked that he supposed so, since Scotland Yard had sent this individual to him on the same principle that some doctors refer their incurable patients to quacks, with the idea that nothing can be done for them anyway, and that whatever happens they are no worse off than they were formerly. Dr. Watson made no reply. One does not like to think that an ethical practitioner would knowingly send any patient to a quack. Holmes probably was in a facetious mood when he made this remark, and was simply teasing Watson, who was not known for his sense of humor.

Dr. Watson clearly recognized a doctor's responsibility to his patient, because in *The Sign of the Four* he writes: "Remember that I speak not only as one comrade to another but as a medical man to one for whose constitution he is to some extent answerable." In *The Dying Detective*, Holmes twitted Watson by reminding him that he was after all only a general practitioner with rather mediocre qualifications. This hurt Dr. Watson, and his comment was that Holmes might be his master elsewhere, but not in the sickroom. Here undeniably Watson was his master.

Like other general practitioners, Dr. Watson undoubtedly had hard days, but presumably he answered night calls. The following occurs in *The Man with the Twisted Lip*:

One night ... there came a ring to my bell, about the hour when a man gives his first yawn and glances at the clock. I sat up in my chair, and my wife ... made a little face of disappointment.... I groaned, for I was newly back from a weary day.

Obviously, Dr. Watson had never had a lucrative practice, for he states: “Everyone was out of town, and I yearned for the glades of the New Forest or the shingle of Southsea. A depleted bank account had caused me to postpone my holiday....” (*The Resident Patient*). In a similar vein, “I have nothing to do today. My practice is never very absorbing” (*The Red-Haired League*). In *A Scandal in Bohemia* occurs a reference which further indicates that Dr. Watson was not a busy practitioner: “At three o’clock precisely I was at Baker Street, but Holmes had not yet returned.... I sat down beside the fire, however, with the intention of awaiting him however long he might be.” This shows that Dr. Watson was not loath to be absent from his practice in the middle of the day for an indeterminate length of time. Apparently, he thought nothing of this, but sat down calmly out of reach of his patients to wait for his friend.

We must not infer, however, that his practice was always dull, for on one occasion he explains that he had pressing professional business of his own, and it was not possible for him to accompany Holmes (*The Adventure of the Illustrious Client*). And again: “A professional case of great gravity was engaging my attention at the time, and the whole of next day I was busy at the bedside of the sufferer” (*The Red-Haired League*). Also, once when Holmes suggested that he and Watson go out of town, the latter writes that although it was easy for Holmes to go almost any time, he (Watson) had to do a certain amount of planning and hurrying because his practice was not inconsiderable. Holmes was cognizant of the fact that Watson had definite obligations, for once Holmes told a client that not only was he a busy man but that Dr. Watson had his patients to consider (*The Adventure of the Creeping Man*).

Apparently, at the time Watson wrote, it was common for physicians to buy and sell their practice. Several references are made to this custom. Watson describes how he had purchased his practice:

I had bought a connection.... Old Mr. ... from whom I purchased it, had at one time an excellent practice, but his age, and an affliction ... had very much thinned it.... Until when I purchased it from him it had sunk from twelve hundred to a little more than three hundred a year.

The Stock Broker's Clerk

It must be remembered that at that time the English pound was worth almost five dollars; so the practice described by Dr. Watson had been bringing in about \$6,000 per year, but had sunk to \$1,500. In those days, of course, money went much further than it does now, and \$6,000 a year was a splendid income.

On another occasion, Watson relates that, at the request of Holmes, who wanted Watson to come to live with him at Baker Street, he sold his small Kensington practice at a rather good figure to a young doctor. A few years later, he found that the purchaser was a distant relative of Holmes, and it was the latter who actually had put up the money (*The Adventure of the Norwood Builder*).

Dr. Watson takes sly digs at specialists. Once he remarks that a certain individual was an excellent antagonist, and that he was as cool as ice, silky voiced and soothing as a fashionable consultant (*The Adventure of the Illustrious Client*). And another time, he states that he read in his friend's eye the arrogance which a learned specialist might experience who had been called to see a case only to find the patient had measles (*The Adventure of the Abbey Grange*).

As to the drugs used in Watson's day, only about a dozen are mentioned. Among them were ammonia, amyl nitrite, brandy, caffeine, ether, chloroform, iodoform, carbolic acid, curare, and silver nitrate. A survey made in England in the early part of this century showed that physicians considered from twenty to twenty-five drugs necessary to practice medicine satisfactorily. If a survey were made today, it is likely that more would be listed, because during the past few years potent antibiotic agents have been discovered and great strides have been made in chemotherapy. Oliver Wendell Holmes, the famous poet-physician, if living today, could not write as he did in the last century: "If the whole materia medica (excepting opium

and ether) as now used could be sunk to the bottom of the sea it would be all the better for mankind—and all the worse for the fishes.”

We may think of the faithful Dr. Watson answering a call, perhaps late at night, his stethoscope concealed in his hat, and his trusty medical bag filled with the acceptable drugs. We can picture him further, riding along in his hansom—the lights of which shine dimly through the fog—rattling over the cobblestones of the old London streets, carrying on the tradition of his noble profession. And we pray that this priceless tradition will continue as long as mankind suffers from illnesses of the mind or body.

FOOTNOTES:

- [1] Cassamajor, Louis, "Brain Fever," *Journal of the American Medical Association*, CXLIX (1952), 1443-46.
- [2] Klein, W. E., and R. H. Bradshaw, "Portuguese Man-of-War Sting," *Armed Forces Medical Journal*, II (March, 1951), 509-512.
- [3] Waite, C. L., "Medical Problems of an Underwater Demolition Team," *Armed Forces Medical Journal*, II (Sept., 1951), 1317-26.
- [4] Sollman, T., *A Manual of Pharmacology* (Philadelphia, W. B. Saunders Co., 6th ed., 1946), p. 104.
- [5] H. T. Webster, "Observations on Sherlock Holmes as an Athlete and Sportsman," *The Baker Street Journal*, III (1948), No. 1, pp. 24-31.
- [6] H. T. Webster, *op. cit.*

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